

Finite tensor savings and exact Fourier circuits

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Abstract

We construct exact nonuniform Fourier circuits of size $o(n \log n)$ along an unbounded sequence of lengths, counting every addition, subtraction, and scalar multiplication. This refutes the $\Omega(n \log n)$ lower bound in the unrestricted complex linear-circuit model. The construction uses a finite tensor saving: a tensor power of some invertible nonmonomial complex matrix can be computed with fewer matrix calls than the standard tensor-axis algorithm on the same coordinates, when invertible monomial maps are allowed freely between calls.

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1 Introduction

The discrete Fourier transform is a basic test case for arithmetic complexity. Its familiar $O(n \log n)$ algorithms make the corresponding lower-bound question particularly sharp: does every exact computation require a constant multiple of $n \log n$ operations? The answer depends on the computational model. We consider complex linear circuits with arbitrary prechosen coefficients and prove that this lower bound is false in its unrestricted nonuniform form.

The argument centers on a finite question about tensor products. For integers $q, b \geq 2$, applying a fixed $q \times q$ matrix on each axis computes its b th tensor power with bq^{b-1} matrix calls. A *monomial matrix* is a permutation matrix times an invertible diagonal matrix. We prove that, for at least one invertible nonmonomial matrix, an exact computation improves this count. A single such finite saving is enough: tensor amplification turns it into an exponent improvement, and a factorization that preserves the number of coordinates transfers the improvement to Fourier matrices. The existence proof develops a second set of tools, based on prices of matrix calls and their behavior under feedback and changes of boundary conditions.

1.1 The model and the result

For $n \geq 2$, put $\zeta_n = \exp(2\pi i/n)$ and let

$$F_n = (\zeta_n^{jk})_{0 \leq j, k < n}. \quad (1.1)$$

A linear circuit has inputs $x_0, \dots, x_{n-1}, 0$ and a finite directed acyclic graph of scalar gates. Each gate computes $u + v$, $u - v$, or λu from previously available values, with a fixed $\lambda \in \mathbb{C}$. Each such gate costs one operation, including every scalar multiplication. Previously computed values can be reused arbitrarily, and the outputs are designated available values. A fixed combination $\alpha u + \beta v$ therefore costs at most three gates.

The graph and coefficients may depend on n , but not on the input. Coefficients have no bound on magnitude, algebraic degree, or description length; their generation is not charged. Depth, storage, cancellation, and conditioning are unrestricted. All n output coordinates must equal $F_n x$ exactly for every $x \in \mathbb{C}^n$. Let $L(n)$ be the minimum number of gates in such a circuit.

Theorem 1.1. *In this model,*

$$\liminf_{n \rightarrow \infty} \frac{L(n)}{n \log_2 n} = 0.$$

Equivalently, for every $c > 0$ and every integer $N_0 \geq 2$, there is an integer $n \geq N_0$ and an exact circuit for F_n with fewer than $cn \log_2 n$ gates.

The $\Omega(n \log n)$ Fourier lower-bound question is discussed in Ailon [1, Abstract] and Alman and Rao [4, Section 1]. In the model above, Theorem 1.1 refutes the assertion that $L(n) \geq cn \log_2 n$ for some fixed $c > 0$ and all sufficiently large n . The proof gives an unbounded sequence of lengths, formed as products of distinct primes, with a vanishing normalized gate count. It is an existence theorem in the stated nonuniform model. The coefficients are chosen separately for each resulting circuit; their preparation is outside this model. The theorem concerns arithmetic operation counts along those lengths and makes no assertion about coefficient size, numerical stability, or bit complexity. Neither a uniform construction nor an all-length upper bound is needed for this quantified conclusion.

1.2 Historical context

The fast Fourier transform of Cooley and Tukey organizes a composite-length transform into smaller transforms and diagonal factors [9]. Coprime factors have a particularly useful structure: Chinese remainder permutations identify the transform with a tensor product without intervening diagonal factors. This is the Good–Thomas factorization; Good’s treatment gives the multidimensional and coprime formulas, and Cooley, Lewis, and Welch describe Thomas’s construction and its relation to Good’s work [11, Sections 11–12] [10, pp. 1675–1677]. Convolution supplies another classical route to general lengths. The 1969 chirp-transform paper of Rabiner, Schafer, and Rader derives the reduction to diagonal scalings and a convolution, crediting Bluestein’s 1968 work for the quadratic-exponent identity [17, Section II]. Bluestein’s journal treatment appeared in 1970 [6]. The transfer proved here uses the coprime tensor structure together with a local factorization on exactly the original coordinates. Preserving that width is what permits a finite tensor saving to survive synchronization.

Lower bounds have been proved under substantial restrictions on the algorithm. Morgenstern’s determinant argument establishes a bound of order $n \log n$ for the unnormalized transform in a bounded-coefficient model [14]. Ailon proves a lower bound for the normalized transform in a model with exactly n registers and unitary 2×2 gates [1, Section 2 and Theorem 2.1] and, in a separate model, relates the operation count to the conditioning of every intermediate transformation [2, Definition 2.1 and Theorem 3.1]. Those hypotheses are significant here: the circuits in Theorem 1.1 are allowed arbitrary coefficients and arbitrary intermediate conditioning. An unrestricted scalar circuit can pass between normalized and unnormalized transforms with $O(n)$ additional gates. Such a scaling need not respect a prescribed coefficient bound, so the normalization must be retained when comparing the restricted models.

Upper bounds have also improved within the $n \log n$ scale. In particular, Alman and Rao obtain new leading constants for power-of-two transforms, measured in real arithmetic operations [4, Theorem 1.2]. A fixed leading-constant improvement does not settle the quantified question addressed here. The essential step in this paper is to turn one strict finite tensor saving into a smaller exponent in the tensor order, and then preserve that saving when passing to an unbounded sequence of Fourier transforms.

Tensor amplification and duality also appear in recent work on Kronecker-power circuits. Alman and Li use asymptotic-spectrum duality to characterize the asymptotic size of depth-two linear

circuits [3, Sections 1.3 and 4]. Their circuits are represented by rank-one decompositions, with size measured by the nonzero entries in the two matrix factors. Our finite-win question concerns sequential invertible matrix calls on a fixed coordinate set. Its price construction must therefore preserve the order of each program while combining different matrix types.

The local Fourier construction uses established ideas from structured linear algebra and reversible computation. Kuznetsov gives an LDL^T factorization of geometric Vandermonde matrices, including Fourier matrices, whose triangular factors are diagonal scalings of triangular Toeplitz matrices [13, Theorem 1]. We give a Newton-polynomial derivation of this identity and implement its factors on the original coordinates. Low displacement rank supplies the convolution representation used in that implementation [12, Lemmas 1–2]. Computing, reading the output, and reversing a computation is the uncomputation pattern of Bennett [5, pp. 526–529]. Restoration from arbitrary workspace contents is achieved by the cleanup construction for transparent programs [8, Section 3]. We prove the particular linear replay identity, its pair-layer bound, and its fit inside the existing coordinate set.

The companion manuscript *An explicit power saving for the exact discrete Fourier transform* gives a quantitative all-length algorithm in a model that also charges scalar preparation and logarithmic-word address operations, with a specified root of unity supplied [15, Theorem 1.1]. The present paper develops the general finite-win existence argument: comparison packing, the resulting price laws, and the reflection contradiction are proved in full independently of that algorithm.

1.3 The mechanism of the proof

For an invertible $q \times q$ matrix A , the usual computation of $A^{\otimes b}$ uses bq^{b-1} calls to A , acting along tensor fibers. Call a strict improvement a *finite win*: an exact word on q^b coordinates, using fewer calls, with permutations and invertible diagonal maps allowed between them. The matrix A must be nonmonomial. This intermediate convention is confined to matrix words. Every scalar multiplication is charged when such a word is implemented as a circuit.

The argument has two parts.

- (i) *A finite win implies Theorem 1.1.* For any nonmonomial A , one fixed positive-call pattern, with adjustable monomial factors, realizes every invertible matrix of the same width. A finite win then gives circuits for $A^{\otimes k}$ with $O(q^k(k+1)^\alpha)$ gates for some fixed $\alpha < 1$. Each Fourier matrix F_r is factored, on its original r coordinates, into $O(1 + \log^4 r)$ layers, each monomial or a disjoint union of invertible two-coordinate maps with any remaining coordinates fixed. The temporary convolution computations borrow arbitrary existing values and restore them by cancellation. Positive generation places the pair maps in fixed-width tuples and replaces them by a common call pattern in A . Synchronizing this pattern across distinct prime factors makes each call slot a direct sum of tensor powers of A . The sector widths sum to the Fourier length, so the amplified bound controls the entire slot. The Chinese Remainder Theorem then gives the desired sequence.
- (ii) *A finite win must exist.* Suppose that none exists. Compare tensor-axis computations with words using a finite collection of matrix types. If several comparisons together save calls to every type, we can place their programs on disjoint tensor sectors and schedule them as calls to one larger block-diagonal matrix. This would give a finite win. Separation and compactness turn the impossibility of such simultaneous savings into one price function p , positive on nonmonomial matrices, with exact tensor and direct-sum laws.

We next ask how prices change when coordinates are eliminated. Split the coordinates of an invertible matrix into data and state, and apply it repeatedly to fresh data while retaining the

state. Repeat this construction on many tensor axes. The cost of correcting the initial state becomes negligible compared with the number of cell calls. This proves that an invertible corner costs no more than its containing matrix. Comparing multiplication modulo different scalar polynomials also permits closing the state connection and specializing rational matrix families. All these conclusions concern exact matrices; they do not assume continuity of the price.

Finally, normalize a nontrivial elementary shear to have price one. A legal exchange of one input basis member with one output basis member preserves price, provided both exchanged lists remain bases. Applying these exchanges to signed bit-coordinate matrices gives a family of involutions, with integer parameter d , width $w = w(d)$, and price dw . The constructed $+1$ eigenspace is the graph of an invertible matrix. From the orthogonal reflection R in that graph, we can recover the involution's price at an additional cost of only $O(w)$. The same subspace is the kernel of a real symmetric matrix with at most $(d/2 + 1)w$ supported off-diagonal pairs. A product of diagonal and two-coordinate perturbations has this sparse matrix as its derivative at zero. At generic parameter values, each dense pair costs at most $3/2$. Feedback adds only $O(w)$, and two algebraic specializations carry the bound to R . Thus the two descriptions give

$$dw - O(w) \leq p(R) \leq \frac{3d}{4}w + O(w),$$

with absolute implied constants. They are incompatible for sufficiently large d .

These parts meet at one precise interface: the finite word of Theorem 2.1. The transfer in Theorem 3.4 works for any word satisfying that definition, while Theorem 8.3 establishes that at least one exists. In particular, the proof does not require a finite win for every nonmonomial matrix.

1.4 Conventions and order of the argument

All matrices and scalar coefficients are over $\mathbb{C}$, except where real symmetry and real orthogonality are explicitly used. Tensor products are ordinary Kronecker products. For an invertible square matrix A , its width is denoted by $|A|$. A matrix call acts on an ordered tuple of distinct coordinates and fixes every other coordinate. Zero-dimensional blocks may be omitted. A superscript T denotes ordinary transpose, not conjugate transpose.

The price function is introduced only under the assumption that there is no finite win. It is an auxiliary real-valued function, not an asserted lower bound for circuit size. Every limiting argument proves an inequality for exact finite matrices. In particular, specialization never replaces an exact circuit by a limit of approximate computations.

Constants in an O bound may depend on fixed local matrices and their dimensions. When a tensor order k and a run length t vary, each polynomial bound in k has a fixed exponent and is independent of t unless stated otherwise. Scalar coefficients may depend on these parameters, as the model permits. We specify the order of choices at the packing, feedback, and specialization steps.

Sections 2 and 3 prove the transfer from one finite win, through positive generation, amplification, and exact-width Fourier layers. Section 4 constructs the price function from the opposite assumption. Sections 5 and 6 establish corner contraction, feedback, and algebraic specialization. Section 7 gives scalar-pivot invariance and the dense-pair bound. Finally, Section 8 uses those laws to force the reflection contradiction and prove Theorem 8.3, which closes the proof of Theorem 1.1.

2 Positive words and tensor amplification

For an invertible square matrix A , write $|A|$ for its width. A *monomial matrix* is a permutation matrix times an invertible diagonal matrix. A *call to A on w coordinates*, where $w \geq |A|$, applies A to an ordered tuple of $|A|$ distinct coordinates and fixes the others. A *matrix word* is a finite composition of such calls and full-width monomial matrices. In a word using one specified matrix A , its *call count* counts only occurrences of A ; monomials have zero call count. This convention concerns words alone. In the scalar circuit model, a monomial on w coordinates costs at most w scalar multiplications, with its permutation realized by designating the appropriate available values.

Definition 2.1 (Finite win). A finite win consists of an invertible nonmonomial matrix $A \in \text{GL}_q(\mathbb{C})$, an integer $b \geq 2$, and a word on exactly q^b coordinates implementing $A^{\otimes b}$ with g calls to A , where

$$g < bq^{b-1}. \quad (2.1)$$

The comparison in Equation (2.1) is with the usual tensor-axis algorithm: for each of the b axes, it makes q^{b-1} calls to A . Nonmonomiality implies $q \geq 2$. It also implies that $A^{\otimes b}$ is nonmonomial: an invertible nonmonomial matrix has a row with at least two nonzero entries, and tensoring that row with any nonzero rows preserves this property. Thus a winning word necessarily has $g \geq 1$.

2.1 One fixed positive pattern generates every target

The following statement supplies both the local substitutions in the Fourier transfer and positive-call identity words used later in packing. The positions of all calls are fixed before the target matrix is chosen.

Lemma 2.2 (Positive generation). *Let $A \in \text{GL}_q(\mathbb{C})$ be nonmonomial. There is an integer $h > 0$ such that every $H \in \text{GL}_q(\mathbb{C})$ admits a factorization*

$$H = M_0 A M_1 A \cdots A M_h, \quad (2.2)$$

with exactly h occurrences of A and monomial matrices M_i . In particular, identity admits such a factorization with a positive number of calls. More generally, for every fixed $w \geq q$, there is a fixed positive pattern on w coordinates realizing every element of $\text{GL}_w(\mathbb{C})$ by embedded calls to A and monomials.

Proof. We build a fixed word whose adjustable diagonal factors give a full-rank differential, then show that its actual values contain a nonempty Zariski open subset of $\text{GL}_q(\mathbb{C})$. Every invertible target is a product of two values from this subset, so doubling the word gives the common pattern.

Let E_{ij} denote the matrix units. Let $\mathfrak{d} \subset \text{Mat}_q(\mathbb{C})$ be the diagonal subspace and let $\mathcal{S}$ be the semigroup of words in A and monomials, including identity. Inverses are not among the specified calls. First we prove

$$\text{span}_{s \in \mathcal{S}} s \mathfrak{d} s^{-1} = \text{Mat}_q(\mathbb{C}). \quad (2.3)$$

If $A \mathfrak{d} A^{-1} \subset \mathfrak{d}$, each rank-one idempotent $A E_{ii} A^{-1}$ would be a diagonal rank-one idempotent, hence some $E_{\pi(i)\pi(i)}$. Orthogonality of these idempotents forces π to be a permutation. The identity $A E_{ii} = E_{\pi(i)\pi(i)} A$ then says that column i of A is supported in row $\pi(i)$; invertibility would make A monomial. Consequently some $X \in A \mathfrak{d} A^{-1}$ has $X_{ij} \neq 0$ for $i \neq j$.

Write $\mathcal{V}$ for the span on the left of Equation (2.3). For every invertible diagonal $D = \text{diag}(d_1, \dots, d_q)$, the matrix $D X D^{-1}$ belongs to $\mathcal{V}$. If a linear functional ℓ vanishes on $\mathcal{V}$, then

$$0 = \ell(D X D^{-1}) = \sum_a X_{aa} \ell(E_{aa}) + \sum_{a \neq c} X_{ac} \ell(E_{ac}) d_a / d_c \quad (d_1, \dots, d_q \in \mathbb{C}^*).$$

The Laurent monomials 1 and d_a/d_c for $a \neq c$ are distinct and linearly independent. Thus $X_{ij}\ell(E_{ij}) = 0$, and so $\ell(E_{ij}) = 0$. Finite-dimensional linear duality gives $E_{ij} \in \mathcal{V}$. Conjugation by any permutation preserves $\mathcal{V}$, because left multiplication of a word by that permutation is again a word. Hence $\mathcal{V}$ contains every off-diagonal matrix unit. It also contains $\mathfrak{d}$, using $s = I_q$. This proves Equation (2.3).

We next arrange the spanning directions along one word. Start with the prefix $h_0 = I_q$ and its diagonal directions. If the directions collected so far span a proper subspace and the current prefix is h , then

$$\text{span}_{s \in \mathcal{S}}(hs)\mathfrak{d}(hs)^{-1} = h \text{Mat}_q(\mathbb{C})h^{-1} = \text{Mat}_q(\mathbb{C}).$$

Choose a word s whose directions enlarge the collected span, and append it to the right of h in written product order. Each choice increases the dimension, so finitely many choices suffice. Denote the successive appended words by $s_1, \dots, s_r$ and their prefixes by $h_i = s_1 \cdots s_i$. The spaces $h_i \mathfrak{d} h_i^{-1}$, including $i = 0$, now span $\text{Mat}_q(\mathbb{C})$.

Insert variable diagonal factors in the product

$$\Phi(D_0, \dots, D_r) = D_0 s_1 D_1 s_2 \cdots s_r D_r.$$

At $D_0 = \cdots = D_r = I_q$, its value is h_r . Varying D_i in the diagonal direction E_{aa} gives the right-translated derivative

$$(\delta\Phi)h_r^{-1} = h_i E_{aa} h_i^{-1}.$$

Thus the derivative has rank $a = q^2$. Choose a diagonal entries whose derivative columns are independent, and fix all other entries to 1. Listing matrix entries produces a polynomial map

$$f : \mathbb{C}^a \longrightarrow \mathbb{C}^a$$

whose Jacobian determinant is nonzero at $(1, \dots, 1)$. Parameters in $(\mathbb{C}^*)^a$ give admissible words. We show next that their image contains an actual nonempty Zariski open subset of $\text{GL}_q(\mathbb{C})$.

Write $x_1, \dots, x_a$ for the selected parameters and $f_1, \dots, f_a$ for the output entries. The f_i are algebraically independent. Indeed, a nonzero polynomial relation $P(f) = 0$ of least total degree would, upon differentiation and inversion of the Jacobian over $\mathbb{C}(x_1, \dots, x_a)$, imply $(\partial P / \partial y_i)(f) = 0$ for every i . In characteristic zero at least one of these is a nonzero polynomial of smaller degree, a contradiction. Both fields in the inclusion

$$\mathbb{C}(f_1, \dots, f_a) \subset \mathbb{C}(x_1, \dots, x_a)$$

therefore have transcendence degree a . The latter is an algebraic extension generated by the finitely many x_i , so it is finite. Each of the finitely many elements x_i, x_i^{-1} satisfies a monic equation over the smaller field. Choose a nonzero polynomial $g(y_1, \dots, y_a)$ divisible by the denominators of all coefficients of these equations. With

$$\begin{aligned} R &= \mathbb{C}[f_1, \dots, f_a, 1/g(f)], \\ B &= R[x_1, \dots, x_a, x_1^{-1}, \dots, x_a^{-1}] \subset \mathbb{C}(x_1, \dots, x_a), \end{aligned}$$

all the displayed generators of B are integral over R . Reducing their powers by their monic equations shows directly that B is a finite R -module. Both are domains and $R \hookrightarrow B$ is injective. This is the finite-type integral-extension criterion [19, Tag 02JJ, Lemma 10.36.5].

Fix any target $y \in \mathbb{C}^a$ with $g(y) \neq 0$, and let $\mathfrak{m} = (f_1 - y_1, \dots, f_a - y_a) \subset R$. Algebraic independence identifies $R/\mathfrak{m}$ with $\mathbb{C}$. Localizing the inclusion above gives $R_{\mathfrak{m}} \hookrightarrow B_{\mathfrak{m}}$ inside the field $\mathbb{C}(x_1, \dots, x_a)$, so $B_{\mathfrak{m}}$ is a nonzero finite module over the local ring $R_{\mathfrak{m}}$. Its residue algebra

$$B_{\mathfrak{m}}/\mathfrak{m}B_{\mathfrak{m}} \tag{2.4}$$

is nonzero. This is the lying-over step for the integral inclusion [19, Tag 00GQ, Lemma 10.36.17]. For completeness, if it were zero, a finite list of module generators $b_1, \dots, b_e$ would satisfy $b_i = \sum_j c_{ij} b_j$ with all c_{ij} in the maximal ideal of R_m . The determinant of $I - (c_{ij})$ is a unit in this local ring. Multiplication by its adjugate forces every b_i to be zero, contradicting $B_m \neq 0$.

The algebra in Equation (2.4) is finite dimensional over $\mathbb{C}$. A quotient by a maximal ideal is a finite field extension of $\mathbb{C}$, and hence is $\mathbb{C}$ itself. The resulting homomorphism $B \rightarrow \mathbb{C}$ sends x_i to some $\xi_i \neq 0$, since x_i^{-1} is also in B . The polynomial identities defining the f_i descend to $f_i(\xi) = y_i$. These are actual admissible parameters of the fixed word. In particular its image contains

$$\Omega = \{Y \in \text{Mat}_q(\mathbb{C}) : g(Y) \det Y \neq 0\},$$

a nonempty open set, because $g(Y) \det Y$ is a nonzero polynomial.

Finally fix $H \in \text{GL}_q(\mathbb{C})$. There is a V for which both $V \in \Omega$ and $V^{-1}H \in \Omega$. To see this without any limiting argument, clear powers of $\det V$ from $g(V^{-1}H)$. Its numerator is a nonzero polynomial: the substitution $V \mapsto V^{-1}H$ is a bijection of $\text{GL}_q(\mathbb{C})$, with inverse $W \mapsto HW^{-1}$, so this rational function is not identically zero. Avoid simultaneously the zeros of that numerator, $g(V)$, and $\det V$. A finite product of nonzero polynomials over $\mathbb{C}$ is nonzero and has a point where it does not vanish. This supplies the required V . Setting $W = V^{-1}H$ gives $H = VW$, with both factors realized by the same fixed word pattern. Two copies of that pattern therefore realize every H .

The initial pattern contains at least one occurrence of A : otherwise its values have one fixed monomial support, a space of dimension q , contradicting the derivative rank $q^2 > q$. Expanding the fixed words s_i and merging adjacent monomials in the doubled pattern gives Equation (2.2) with a fixed $h > 0$. All inverses used in the argument occurred in matrix identities or parameter selection; every factor of the constructed words is a forward call to A or a monomial. For $w > q$, applying the same result to the nonmonomial matrix $A \oplus I_{w-q}$ proves the wider-width assertion, because each call to this matrix is one embedded call to A . $\square$

2.2 Amplifying a finite saving

The next lemma concerns scalar gates, including the cost of every monomial scaling. All its constants are allowed to depend on the single fixed finite win.

Lemma 2.3 (Tensor amplification). *If (A, b, g) and a specified winning word form a finite win, then there are constants $C < \infty$ and $0 < \alpha < 1$ such that $A^{\otimes k}$ has an exact scalar linear circuit with at most $Cq^k(k+1)^\alpha$ gates for every integer $k \geq 0$.*

Proof. Let $L_A(k)$ be the minimum scalar gate count for $A^{\otimes k}$, and put $f(k) = L_A(k)/q^k$. Here $A^{\otimes 0} = I_1$ and $f(0) = 0$. These minima exist: entrywise multiplication and addition give finite circuits, and a nonempty set of nonnegative integer gate counts has a minimum. Set

$$Q = q^b, \quad p = q/Q, \quad j = \lfloor k/b \rfloor.$$

Tensor the winning word with itself over j groups, multiplying the corresponding slots together. The product of these slots is $(A^{\otimes b})^{\otimes j} = A^{\otimes bj}$.

A monomial slot remains monomial after taking this tensor power, and therefore costs at most Q^j scalar gates. Each call slot is permutation-conjugate to $A \oplus I_{Q-q}$. Its j -fold tensor power, after reordering coordinates, has the following exact direct-sum decomposition:

$$(A \oplus I_{Q-q})^{\otimes j} \cong \bigoplus_{r=0}^j (A^{\otimes r})^{\oplus \binom{j}{r}} (Q-q)^{j-r}. \quad (2.5)$$

For a chosen subset of r active factors, the inactive factors supply $(Q - q)^{j-r}$ independent scalar coordinate choices, explaining the multiplicity. The total width is $\sum_r \binom{j}{r} q^r (Q - q)^{j-r} = Q^j$. Computing the blocks separately with minimum circuits gives normalized cost at most

$$\begin{aligned} \frac{1}{Q^j} \sum_{r=0}^j \binom{j}{r} (Q - q)^{j-r} L_A(r) &= \sum_{r=0}^j \binom{j}{r} p^r (1 - p)^{j-r} f(r) \\ &= \mathbb{E} f(K), \quad K \sim \text{Bin}(j, p). \end{aligned} \quad (2.6)$$

In particular, this distribution weights sectors by their coordinate dimensions; the sectors have different widths.

Replicate the resulting circuit over the coordinates of the remaining $k - bj < b$ tensor axes. The normalized cost is unchanged by this replication. If a fixed circuit for A has ℓ_A gates, applying A along each remaining axis costs $\ell_A q^{k-1}$ gates per axis. Since the number of monomial slots in the winning word is fixed, there is a constant C_0 independent of k such that

$$f(k) \leq C_0 + g \mathbb{E} f(K), \quad K \sim \text{Bin}(\lfloor k/b \rfloor, q/Q). \quad (2.7)$$

For $j = 0$ the grouped part is identity and the same inequality follows from the bound on the fewer than b remaining axes. Every permutation used in deriving Equation (2.5) is only a reindexing; every diagonal scaling has already been charged.

Put $\rho = q/(Qb)$. The strict saving gives $g\rho < 1$, and hence we can choose $0 < \alpha < 1$ sufficiently close to 1 that $g\rho^\alpha < 1$. Choose also a number γ with $g\rho^\alpha < \gamma < 1$. Since

$$g \left(\frac{\rho k + 1}{k + 1} \right)^\alpha \longrightarrow g\rho^\alpha,$$

there is $k_0 \geq 1$ such that this expression is at most γ for all $k \geq k_0$. Choose C at least $C_0/(1 - \gamma)$ and large enough that $f(k) \leq C(k + 1)^\alpha$ for $k < k_0$. For $k \geq k_0$, every value of K satisfies $K \leq \lfloor k/b \rfloor < k$. Strong induction, concavity of $x \mapsto x^\alpha$, and $\mathbb{E}K = p\lfloor k/b \rfloor \leq \rho k$ therefore give

$$f(k) \leq C_0 + gC \mathbb{E}(K + 1)^\alpha \leq C_0 + gC(\rho k + 1)^\alpha \leq C_0 + C\gamma(k + 1)^\alpha \leq C(k + 1)^\alpha.$$

All circuits used in this induction compute exact matrices; the expectation is merely the finite weighted sum in Equation (2.6). Multiplying by q^k proves the scalar gate bound

$$L_A(k) = O(q^k (k + 1)^\alpha), \quad 0 < \alpha < 1.$$

□

3 From a finite win to exact Fourier circuits

The amplification in Theorem 2.3 concerns tensor powers of one fixed matrix. To use it for Fourier transforms of different widths, we first construct shallow matrix words without enlarging those widths. A *pair layer* is a direct sum, after a coordinate permutation, of invertible two-coordinate matrices and one-coordinate identities. A *monomial layer* is an invertible monomial matrix on the whole coordinate set. During a recursive construction we also permit a *mixed round*: on disjoint regions it performs monomial layers or pair layers. Such a round splits into at most two layers of the stated types, since the operations on different regions commute.

3.1 Convolutions and borrowed coordinates

We will use the following elementary circuit fact. A convolution with a fixed vector, including truncations and reversals of its inputs or outputs, has a linear DAG with

$$O(v \log(2 + v)) \text{ gates}, \quad O(\log(2 + v)) \text{ depth},$$

where v is the sum of the input and kernel lengths. Moreover its variable inputs and gates have bounded fan-out, including uses at the outputs, with an absolute bound. Here and below depth counts scalar arithmetic gates.

For nonempty vectors of lengths $a, e \geq 1$, pad both by zeros to the least power of two $L \geq a + e - 1$; then $L < 2(a + e - 1)$. Empty operands give the zero map and may be omitted. At $L = 1$ the Fourier maps are identities. At the L th roots of unity, ordinary convolution becomes pointwise multiplication; the transform of the fixed vector is part of the prechosen constants. The binary Fourier recursion transforms even and odd positions separately and combines each pair of transformed entries a, b as $a + \omega b, a - \omega b$. It uses at most three scalar gates per pair, $O(L \log(2L))$ gates in total, and $O(\log(2L))$ depth. Every recursion input feeds one leaf; each transformed entry is used in only its own pair of outputs at the next level. The inverse uses inverse roots and a final scaling by L^{-1} , with the same bounds. Padding zeros require no variable input. The intervening diagonal multiplications and the output selections preserve bounded fan-out. This also proves the same estimates for a fixed number of such convolutions, compositions, and sums. All scalar multiplications in these estimates are charged.

The padded DAG in this argument need not itself be in place. The next lemma explains how its workspace can be supplied by existing coordinates carrying arbitrary values. The forward, output, and reverse sweeps follow the uncomputation pattern of Bennett [5]. The arbitrary initial values also connect this construction with transparent computation [8, Section 3]. The second replay below removes their contribution and supplies the precise shear and layer counts used here.

Lemma 3.1 (Dirty-coordinate replay). *Let $M : \mathbb{C}^e \rightarrow \mathbb{C}^a$ have a linear DAG with ℓ gates and depth H . Suppose every variable input and gate has fan-out at most δ , counting designated output uses. On disjoint source coordinates x , target coordinates y , and ℓ further coordinates z , there is a word of coordinate shears realizing*

$$(x, y, z) \mapsto (x, y + Mx, z).$$

It uses at most $8\ell + 2a$ shears and $O((\delta + 1)(H + 1))$ pair layers. The initial values of z are arbitrary, and may depend on x, y or other data.

Proof. Assign one distinct coordinate z_v to each gate v , ordered topologically. To simulate the gate, add its prescribed linear combination of predecessors into z_v . A scalar gate requires one shear, and an addition or subtraction requires at most two. An edge from the literal input zero is omitted. Predecessors are either original source coordinates or earlier gate coordinates; in particular none is the coordinate being updated. The forward sweep therefore has the form

$$z \mapsto Uz + Tx, \tag{3.1}$$

where U is unit lower triangular in the chosen order. Write the designated output reads as $Fz_{\text{current}} + Dx$; this includes outputs that are original inputs, and a zero output contributes no read. Each nonzero output is one designated read. Starting with zero gate coordinates computes the original DAG, so

$$FT + D = M. \tag{3.2}$$

Run the forward sweep, add its outputs into y , and undo every shear of that sweep in reverse order. The net effect on y is addition of

$$F(Uz + Tx) + Dx = Mx + FUz,$$

and every scratch coordinate returns to its initial value. Next run the same forward sweep with *all* edges from original source coordinates omitted. Omit the direct-source output term Dx as well. This sweep sends z to Uz . Subtract its outputs FUz from y and undo the sweep. This leaves exactly $y + Mx$ and restores z once more. The target coordinates are never read by either sweep. These identities hold for independent formal variables x, y, z , hence remain valid after substituting correlated values for the initial scratch.

To obtain the layer bound, assign gates their longest-path depth levels. Within a level, all destinations are new gate coordinates and all predecessors belong to earlier levels or to the sources. The undirected multigraph of shear edges has maximum degree at most $\Delta = \max\{2, \delta\}$: a destination has at most two incoming edges and a predecessor has at most δ uses. Repeated edges can be combined, or retained with their bounded multiplicity. Greedy edge coloring uses at most $2\Delta - 1$ colors, because an edge meets at most $2\Delta - 2$ other edges. Each color is a disjoint-pair layer. These shears can be reordered by colors because no destination in this level is a predecessor in the same level. Output additions have the same bounded-degree property on source or gate coordinates and target coordinates. Reversing the colored levels gives the exact inverse sweep. Thus four sweeps and two output rounds need at most $4(2\Delta - 1)H + 2(2\Delta - 1)$ layers. Each sweep uses at most 2ℓ shears and each output round at most a , proving the count. $\square$

3.2 An exact-width factorization

The structured-matrix step uses the displacement-rank method of Kailath, Kung, and Morf [12, Lemmas 1–2]: a matrix whose difference from a shifted copy has low rank can be reconstructed from convolution factors. We give the rectangular reconstruction in the proof, then place its workspace inside the original coordinate set.

Lemma 3.2 (Triangular Toeplitz layers). *Every invertible lower triangular Toeplitz matrix of width r has a factorization on exactly r coordinates into $O(1 + \log^4 r)$ monomial and pair layers. The implied constant is absolute and independent of its entries.*

Proof. Write $T_r = (h_{i-j})_{0 \leq j \leq i < r}$, with zero entries above the diagonal and $h_0 \neq 0$. Let $g_0, g_1, \dots$ be the coefficients of the inverse formal series to $\sum_{j \geq 0} h_j z^j$, extending the specified h_j arbitrarily beyond those needed. Split at $s = \lfloor r/2 \rfloor$, and put

$$T_r = \begin{pmatrix} T_s & 0 \\ B & T_{r-s} \end{pmatrix}.$$

First apply its two diagonal blocks recursively and in parallel. To finish, add from the first block into the second by $C = BT_s^{-1}$. With global row and column indices its entries are

$$C_{ij} = \sum_{\nu=j}^{s-1} h_{i-\nu} g_{\nu-j}, \quad s \leq i < r, \quad 0 \leq j < s. \quad (3.3)$$

Partition the source and target ranges into consecutive chunks, of a size to be specified below, and consider a chunk matrix M of size $a \times e$. Let S_a, S_e be lower-shift matrices, with ones in

positions $(u, u-1)$. For entries away from the first row and first column of the chunk, Equation (3.3) gives

$$M_{uv} - (S_a M S_e^\top)_{uv} = C_{ij} - C_{i-1, j-1} = -h_{i-s} g_{s-j}.$$

The right side is an outer product on those entries. Extending this outer product to the whole chunk leaves only its first row and first column to correct; these have rank at most one each. Consequently

$$E = M - S_a M S_e^\top = \sum_{\nu=1}^{\rho} v^{(\nu)} (w^{(\nu)})^\top, \quad \rho \leq 3. \quad (3.4)$$

Nilpotence of the shifts gives the terminating telescoping identity

$$M = \sum_{l \geq 0} S_a^l E (S_e^\top)^l.$$

For one outer product $vw^\top$ its summand matrix has entries $\sum_{l \geq 0} v_{u-l} w_{j-l}$, with both vectors zero-extended. It applies to $x \in \mathbb{C}^e$ by the two maps

$$b_l = \sum_{j=l}^{e-1} w_{j-l} x_j \quad (0 \leq l < e), \quad c_u = \sum_{l=0}^{e-1} v_{u-l} b_l \quad (0 \leq u < a). \quad (3.5)$$

The first is a convolution after reversing the input and selecting coefficients; the second is a truncated convolution. Sum the at most three resulting vectors. The preceding binary-FFT construction therefore supplies a DAG for Mx satisfying, for absolute constants C_0, C_1, δ_0 ,

$$\ell \leq C_0(a+e) \log_2(2+a+e), \quad H \leq C_1 \log_2(2+a+e), \quad \delta \leq \delta_0. \quad (3.6)$$

These bounds include the sums of the at most three terms and all output uses. In particular, copying the input to these three fixed networks multiplies its fan-out by only a constant. The constants do not depend on the Toeplitz entries or the chosen outer products.

Choose once and for all $D \geq 16C_0 + 16$ and, for sufficiently large r , take chunk size

$$h = \left\lfloor \frac{r}{D \log_2 r} \right\rfloor.$$

Choose a fixed threshold r_0 so that, for all $r \geq r_0$, $h \geq r/(2D \log_2 r) \geq 1$. The final chunk in either range may be shorter. For every chunk pair,

$$a+e \leq \frac{2r}{D \log_2 r} \leq \frac{r}{8}, \quad \log_2(2+a+e) \leq 2 \log_2 r, \quad \ell \leq \frac{4C_0 r}{D} < \frac{r}{4}.$$

At least $7r/8$ coordinates of this very recursion block lie outside the source and target chunks. They suffice for the ℓ scratch coordinates in Theorem 3.1. No extra coordinate is introduced for the padded FFT: its zero inputs require no register, and every gate requiring a register is already counted in ℓ . The lemma implements addition of this chunk's contribution in $O(\log r)$ pair layers and restores every other coordinate.

Process the chunk pairs serially. Source chunks remain unchanged, target additions accumulate, and coordinates in other chunks can be borrowed again with their current values, even if already updated. There are at most $r/h + 2 = O(\log r)$ chunks in total and hence $O(\log^2 r)$ pairs. The cross-block update needs $O(\log^3 r)$ layers. For $r < r_0$, elimination expresses T_r using scalings and coordinate shears on the same coordinates, at a bounded cost depending only on the fixed r_0 .

Let $d(r)$ denote the worst mixed-round depth of this construction. The child recursions use only their own disjoint blocks, so their depths take a maximum, and

$$d(r) \leq \max\{d(\lfloor r/2 \rfloor), d(\lceil r/2 \rceil)\} + O(1 + \log^3 r).$$

Summing along the $O(\log r)$ balanced recursion levels gives $d(r) = O(1 + \log^4 r)$. Splitting mixed rounds into at most two layers finishes the proof, still on exactly r coordinates. $\square$

The Fourier-to-Toeplitz identity used next is the geometric-Vandermonde $LDL^\top$ factorization of Kuznetsov [13, Theorem 1], specialized to a primitive root of unity. For the general Vandermonde factorization through Newton interpolation, see also Oruç and Phillips [16, Theorem 2.1]. We derive the identity here to fix our conventions, then apply the preceding layer construction to its triangular factors.

Lemma 3.3 (Exact-width Fourier layers). *For every integer $r \geq 1$, the unnormalized Fourier matrix F_r has a factorization on exactly r coordinates into $O(1 + \log^4 r)$ monomial and pair layers, with an absolute implied constant.*

Proof. The case $r = 1$ is identity. For $r \geq 2$ put $u = \zeta_r$, and let P have as its k th column the coefficients of the monic Newton polynomial $\prod_{j=0}^{k-1} (X - u^j)$, for $0 \leq k < r$. Thus P is unit upper triangular. Evaluating these polynomials gives $N = F_r P$, a lower triangular matrix, with

$$N_{ik} = \prod_{j=0}^{k-1} (u^i - u^j) = (-1)^k u^{k(k-1)/2} \frac{H_i}{H_{i-k}} \quad (i \geq k), \quad H_j = \prod_{e=1}^j (1 - u^e), \quad H_0 = 1. \quad (3.7)$$

Only $H_0, \dots, H_{r-1}$ occur, all nonzero since u is a primitive r th root. In particular N is invertible and

$$N = \text{diag}(H_i) T \text{diag}((-1)^k u^{k(k-1)/2}), \quad T_{ik} = \begin{cases} 1/H_{i-k}, & i \geq k, \\ 0, & i < k. \end{cases}$$

The middle matrix is invertible triangular Toeplitz, and the outer factors are invertible diagonal matrices.

Let $D_N = \text{diag}(N_{00}, \dots, N_{r-1, r-1})$ and $L = ND_N^{-1}$. Then $F_r = LD_N P^{-1}$ is a unit-lower/diagonal/unit-upper factorization. Such a factorization, when it exists, is unique: on equating two, a unit lower triangular matrix equals an upper triangular matrix, hence is identity, after which the diagonal and upper factors agree. Since $F_r = F_r^\top$, transposition supplies another such factorization. Uniqueness gives $P^{-1} = L^\top$, and therefore

$$F_r = ND_N^{-1} N^\top. \quad (3.8)$$

Apply Theorem 3.2 to T and include its two diagonal factors to factor N . Transposing a product reverses its factor order; the transpose of each monomial or pair layer is again of that type. Thus Equation (3.8) has the asserted factorization. $\square$

3.3 Prime factors and synchronized generation

The Chinese remainder tensor identity used below is the classical coprime-factor Fourier decomposition [11, Section 12] [10, Equations (6)–(10)]. We include the index calculation to specify the root powers and the coordinate permutations needed for synchronization.

Proposition 3.4 (Transfer of a finite win). *If a finite win exists, there are integers $n_m \rightarrow \infty$ and exact linear circuits for F_{n_m} whose scalar gate counts satisfy*

$$L(n_m) = o(n_m \log_2 n_m).$$

In particular the conclusion is in the model charging additions, subtractions, and multiplication by every prechosen complex scalar.

Proof. Fix the winning matrix A of width q , the winning tensor exponent, and the entire winning word. Since A is nonmonomial, $q \geq 2$. By Theorem 2.3, there are fixed constants C and $0 < \alpha < 1$ such that $A^{\otimes k}$ has a scalar circuit of size at most $Cq^k(k+1)^\alpha$ for every $k \geq 0$. All constants depending on this win remain fixed from now on.

We first obtain enough moderately sized, distinct primes by an elementary estimate. If $\pi(v)$ counts primes at most v , then

$$\binom{2h}{h} \geq \frac{4^h}{2h+1}, \quad \binom{2h}{h} \leq (2h)^{\pi(2h)}.$$

The first inequality follows because the central binomial coefficient is the largest of the $2h+1$ coefficients whose sum is 4^h . For the second, the exponent of a prime p in that coefficient is

$$\sum_{j \geq 1} \left(\left\lfloor \frac{2h}{p^j} \right\rfloor - 2 \left\lfloor \frac{h}{p^j} \right\rfloor \right) \leq \lfloor \log_p(2h) \rfloor.$$

Here each term is zero or one, and the formula itself follows by counting the multiples of $p, p^2, \dots$ in a factorial. The total power contributed by each prime is consequently at most $2h$. It follows that

$$\pi(2h) \geq \frac{h \log 4 - \log(2h+1)}{\log(2h)}.$$

With $h = m^2$, this exceeds $m + \pi(2q)$ for all sufficiently large m . Let $r_1 < r_2 < \dots$ be the primes greater than $2q$ and put $n_m = \prod_{i=1}^m r_i$. Thus $r_m \leq 2m^2$ eventually, $n_m \rightarrow \infty$, and $\log n_m \geq m \log 2$.

For clarity, the Chinese remainder factorization here involves only permutations. Write $n = n_m$ and choose integers e_i whose residue is one modulo r_i and zero modulo every other r_j . These satisfy $e_i^2 = e_i$, $e_i e_j = 0$ for $i \neq j$, and $\sum_i e_i = 1$, all modulo n . The bijections $j = \sum_i e_i j_i$, $k = \sum_i e_i k_i$ yield

$$\zeta_n^{jk} = \prod_{i=1}^m (\zeta_n^{e_i})^{j_i k_i}.$$

For example, take $e_i = (n/r_i)d_i$, where $(n/r_i)d_i \equiv 1 \pmod{r_i}$; then $\zeta_n^{e_i} = \zeta_{r_i}^{d_i}$ and d_i is a unit modulo r_i . Permuting k_i by this unit converts the corresponding factor to F_{r_i} . Hence F_n is $\bigotimes_{i=1}^m F_{r_i}$ up to input and output permutations.

By Theorem 3.3, each factor has at most

$$T = O(1 + \log^4 m)$$

layers, uniformly for these r_i . Pad the shorter sequences in time by identity layers. We next refine a synchronized time step by a bounded number of slots depending only on A .

First reserve a monomial slot: on an axis whose current layer is monomial perform it, and on the other axes perform identity. On a pair-layer axis of width r_i , there are at most $\lfloor r_i/2 \rfloor$ pairs. Split them into at most q batches of at most $\lfloor r_i/q \rfloor$ pairs each. Indeed $\lfloor r_i/q \rfloor \geq r_i/(2q)$, so the

required number of batches is at most q . Within a batch with t pairs, choose a disjoint q -tuple containing each pair: the $2t$ pair coordinates are already distinct and there are enough remaining coordinates because $qt \leq r_i$. The desired map on that tuple is its prescribed pair operation together with identity on the $q - 2$ spectators.

By Theorem 2.2, every such map has an implementation using the same fixed sequence of positive A -call slots and monomial slots; denote its total number of slots by s_A . Run these words in parallel on the disjoint tuples. At a monomial slot their disjoint union, with identity on unused coordinates, is monomial. At an A -call slot the layer is a disjoint union of embedded A calls and identity on unused coordinates. Use the same slots for every axis, padding to q batches and using identity throughout empty batches or axes. The number of refined slots per original time is at most $1 + qs_A$, independently of m and the r_i .

The spectator coordinates may belong to pairs processed in later batches. A completed tuple word is exactly the required pair map direct-sum identity, so every spectator is restored before the next batch on its axis. For simultaneous execution on different axes, the exact product identity

$$\left(\bigotimes_i W_{i,s}\right) \cdots \left(\bigotimes_i W_{i,1}\right) = \bigotimes_i (W_{i,s} \cdots W_{i,1}) \quad (3.9)$$

proves correctness even though intermediate spectator values change. We implement each tensor slot as a complete linear map; no assumption about intermediate values on other axes is required. Throughout, the coordinate width remains exactly n .

It remains to charge these refined slots in the scalar model. A tensor product of monomial layers is itself monomial and costs at most n scalar multiplications; permutations merely reindex available values. At an A -call slot, decompose each axis into its active q -tuples and its unused singleton coordinates. Distributing the tensor product over these blocks expresses that slot, up to permutation, as a direct sum of matrices $A^{\otimes K}$ with $0 \leq K \leq m$. A sector of width q^K costs at most $Cq^K(K+1)^\alpha \leq Cq^K(m+1)^\alpha$. The sector widths sum to n , so the whole slot costs at most $Cn(m+1)^\alpha$. The $K = 0$ sectors are identities and need no gates.

Thus the complete circuit, including all monomial scalings, has size

$$O(n_m(1 + \log^4 m)(m+1)^\alpha). \quad (3.10)$$

The constants in tensor monomials, Fourier convolutions, and generation words may all be prechosen, as permitted by the nonuniform model. A shear uses at most a scalar multiplication and an addition, and each binary fixed-coefficient combination uses at most three charged gates; no unbounded-fan-in operation is used. Finally, division of Equation (3.10) by $n_m \log_2 n_m$ gives $O((1 + \log^4 m)(m+1)^\alpha/m) \rightarrow 0$. All matrix identities and all circuits used above are exact. $\square$

4 Packing finite comparisons into global prices

Assume throughout this section that no finite win in the sense of Theorem 2.1 exists. We construct a function on all invertible complex matrices. Its values measure the costs of matrix calls in an auxiliary argument; they do not make scalar multiplications free in the circuit model.

Theorem 4.1 (Global prices under the no-win assumption). *There is a function*

$$p : \bigcup_{n \geq 1} \text{GL}_n(\mathbb{C}) \longrightarrow [0, \infty)$$

which vanishes exactly on the monomial matrices and has the following properties. For invertible matrices of the indicated sizes,

$$p(LAR) = p(A) \quad \text{if } L, R \text{ are monomial of width } |A|, \quad (4.1)$$

$$p(AB) \leq p(A) + p(B) \quad \text{if } |A| = |B|, \quad (4.2)$$

$$p(A \otimes B) = |B|p(A) + |A|p(B), \quad (4.3)$$

$$p(A \oplus B) = p(A) + p(B), \quad (4.4)$$

$$p(A^{-1}) = p(A^\top) = p(A). \quad (4.5)$$

Moreover,

$$p(U) = 1, \quad U = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}. \quad (4.6)$$

The proof first constructs prices satisfying every finite tensor-word comparison. Symmetrization at the end preserves the laws displayed in the theorem; preservation of the entire original comparison family after that averaging will not be needed.

4.1 The comparison inequalities

Take any tensor target $C_1 \otimes \cdots \otimes C_a$, with $a \geq 1$ and each C_j invertible, and any word implementing it on exactly $Q = \prod_j |C_j|$ coordinates. Include in a finite list $A_1, \dots, A_m$ every distinct nonmonomial matrix occurring either as a factor or as a call; additional nonmonomial types may also be included. Put $q_i = |A_i|$. If n_i factors equal A_i and the word uses g_i calls to A_i , define

$$\Delta_i = n_i Q / q_i - g_i. \quad (4.7)$$

Here $n_i Q / q_i$ is the number of calls in the usual computation along successive tensor axes. Monomial factors and operations have no coordinate in this vector. The desired comparison is

$$\sum_{i=1}^m \Delta_i p(A_i) \leq 0. \quad (4.8)$$

There is no bound on the widths, number of factors, or lengths of words used to define this family of inequalities. Each individual inequality involves finitely many types.

Lemma 4.2 (Packing comparisons). *Let $\Delta^{(1)}, \dots, \Delta^{(R)} \in \mathbb{R}^m$ be any finite family of comparison vectors, with $m \geq 1$ and the type list containing all their nonmonomial factors and calls. Under the no-win assumption there are no numbers $\lambda_r \geq 0$ such that*

$$s_i := \sum_{r=1}^R \lambda_r \Delta_i^{(r)} > 0 \quad \text{for every } 1 \leq i \leq m.$$

Proof. Suppose such a combination exists. We will choose a block-diagonal matrix J containing many copies of every A_i and many identity blocks, and compute $J^{\otimes b}$ in fewer than the usual $b|J|^{b-1}$ calls to J . Its tensor sectors give disjoint places to run copies of the comparison programs. The strict

savings in every type leave room to spread these programs over fewer time slots while preserving each program's order.

For a slot to be one call to J , however, it must contain exactly the prescribed number of calls to each A_i . Random start times will first keep all slot loads below these capacities. We then fill every vacancy with a call belonging to an identity word on separate coordinates. The construction must therefore provide both slack in each slot and enough unused coordinates for these identity words. We first choose block proportions that reserve enough identity space; the strict comparison savings will then supply the slot slack.

Fixed words, species, and proportions. Discard terms with coefficient zero. A comparison whose target has only monomial factors has $\Delta_i^{(r)} = -g_i \leq 0$ for every i . Discarding these terms only increases the numbers s_i , so they may also be discarded. At least one comparison remains. Replace every monomial target factor by an identity of the same width. To obtain its new implementing word, append the inverse of that monomial along the appropriate tensor axis. The appended operation on the full sector is monomial, so all nonmonomial call counts are unchanged. Append width-one identity factors to make every target have the same number $b \geq 2$ of factors. This changes neither its width nor its comparison vector. Keep the entire original type list, including types appearing only in some implementing words. A type absent from every retained target would have only nonpositive retained differences, contrary to $s_i > 0$.

There are now finitely many block species: one species for each A_i , and one identity species I_d for every required identity dimension d , including $d = 1$. Denote this finite dimension set by $\mathcal{D}$. By Theorem 2.2, for every i there is a fixed word on q_i coordinates implementing identity using exactly $h_i > 0$ calls to A_i and monomials. Fix all these words and put $H = \text{lcm}(h_1, \dots, h_m)$.

Choose positive integers a_i , each divisible by H , and positive integers u_d , $d \in \mathcal{D}$. By increasing u_1 , arrange that, for

$$w_0 = \sum_i q_i a_i + \sum_{d \in \mathcal{D}} d u_d, \quad \theta = \frac{\sum_i q_i a_i}{w_0},$$

one has

$$b\theta < (1 - \theta)^b. \quad (4.9)$$

This is possible since θ tends to zero as u_1 increases. For each positive integer v , set

$$w = vw_0, \quad k_i = va_i, \quad \ell_d = vu_d, \quad J = \bigoplus_i A_i^{\oplus k_i} \oplus \bigoplus_{d \in \mathcal{D}} I_d^{\oplus \ell_d}.$$

Thus $|J| = w$. The fixed copy ratios $\rho_i = k_i/w = a_i/w_0$ and $\gamma_d = \ell_d/w = u_d/w_0$ are all strictly positive. We let w tend to infinity only along these multiples of w_0 . In particular every k_i is divisible by H , and the active coordinate fraction is exactly $\sum_i q_i \rho_i = \theta$.

The block decomposition of $J^{\otimes b}$ has one sector for every choice of a block copy on each of its b axes. A sector with species pattern $\sigma = (\sigma_1, \dots, \sigma_b)$ has width equal to the product of those species' widths. Write $\tau_{A_i} = \rho_i$ and $\tau_{I_d} = \gamma_d$. The exact number of sectors with that ordered pattern is

$$\prod_{j=1}^b (w \tau_{\sigma_j}) = \beta_\sigma w^b, \quad \beta_\sigma = \prod_{j=1}^b \tau_{\sigma_j} > 0. \quad (4.10)$$

Repeated species give ordinary powers in this product: choices on distinct tensor axes are independent, so the same block-copy label can occur on two different axes. Patterns differing in any species have disjoint sector sets.

Disjoint replacements and a fixed saving. Let $\sigma(r)$ be comparison r 's ordered target pattern. Choose a fixed $\eta > 0$ sufficiently small that

$$\eta \sum_{r: \sigma(r)=\sigma} \lambda_r \leq \frac{1}{2} \beta_\sigma \quad \text{for every used pattern } \sigma, \quad \eta s_i \leq \frac{1}{2} b \rho_i \quad (1 \leq i \leq m). \quad (4.11)$$

All quantities on the right have already been fixed and are positive. For comparison r , select

$$m_r(w) = H \left\lfloor \frac{\eta \lambda_r w^b}{H} \right\rfloor$$

sectors of pattern $\sigma(r)$. Equation (4.11) ensures that all selections can be disjoint, including selections of different comparisons having the same pattern. Run the comparison's word on each selected sector. On every other sector run the standard axis program. Sectors containing only identity species require no operation, and no such sector was selected.

The standard sector programs would use $G_i^0 = b k_i w^{b-1}$ calls to A_i : choose the axis, one of its k_i copies of A_i , and one coordinate on each remaining axis. The programs after replacement use

$$\begin{aligned} G_i &= b k_i w^{b-1} - \sum_r m_r(w) \Delta_i^{(r)} \\ &= (b \rho_i - \xi_i) w^b + E_i(w), \quad \xi_i = \eta s_i > 0, \end{aligned} \quad (4.12)$$

where $|E_i(w)| \leq H \sum_r |\Delta_i^{(r)}|$. In particular G_i is a nonnegative integer divisible by H . There are only finitely many possible standard sector patterns and replacement words. Hence there is a fixed integer $L \geq 1$ bounding the number of nonmonomial calls in every such program, independently of w . Monomials before, between, and after these calls are retained as part of the program.

Choose a fixed $\varepsilon > 0$ such that

$$0 < \varepsilon < \min(1, b/2), \quad \varepsilon \rho_i < \xi_i/4 \quad (1 \leq i \leq m),$$

and set

$$T = \lfloor (b - \varepsilon) w^{b-1} \rfloor, \quad F = T - L + 1.$$

For all sufficiently large w , $F > 0$ and

$$k_i - \frac{G_i}{F} = \frac{\xi_i - \varepsilon \rho_i}{b - \varepsilon} w + O(w^{2-b}).$$

The constants in the error terms are independent of w . Fix a number $\kappa > 0$ smaller than half the minimum of the positive coefficients in this display. Increasing the lower threshold on w if necessary gives simultaneously

$$\frac{G_i}{F} \leq k_i - \kappa w \quad (1 \leq i \leq m). \quad (4.13)$$

Thus the finite comparisons, their words, b, h_i, H , the species proportions (and θ), η, ξ_i, L , and finally ε, κ have all been fixed before w is taken large.

A schedule respecting all dependencies. Independently for each sector program having at least one nonmonomial call, choose a start uniformly from $\{1, \dots, F\}$. Put its successive nonmonomial calls in consecutive slots starting there. Every call lies in $\{1, \dots, T\}$, and its internal order is preserved. For a fixed type i and slot t , let $X_{i,t}$ be the number of scheduled calls of that type. A given sector contributes an indicator, even if its program uses A_i repeatedly: different positions

in that program require different starts to fall at slot t , and these events are mutually exclusive. Its contribution has mean at most its number of A_i occurrences divided by F . Different sectors have independent starts. Consequently $X_{i,t}$ is a sum of independent Bernoulli variables with $\mathbb{E}X_{i,t} \leq G_i/F$.

For completeness, write $\rho_{\max} = \max_i \rho_i$, and choose $0 < a \leq 1$ so small that $(e^a - 1 - a)\rho_{\max} \leq a\kappa/2$. Such a choice follows, for example, from $e^a - 1 - a \leq ea^2/2$ on $[0, 1]$. Independence, $1 + x \leq e^x$, and Markov's inequality give

$$\begin{aligned} \Pr(X_{i,t} > k_i) &\leq e^{-ak_i} \mathbb{E}e^{aX_{i,t}} \\ &\leq \exp(-ak_i + (e^a - 1)(k_i - \kappa w)) \\ &\leq \exp(-a\kappa w/2). \end{aligned}$$

There are $mT = O(w^{b-1})$ pairs (i, t) , with fixed m, b . The union bound therefore tends to zero. For a sufficiently large admissible w , fix one schedule for which $X_{i,t} \leq k_i$ for every type and every slot. Independence between different slots is neither claimed nor needed.

Completing the vacancies with identities. For type i , the number of unfilled call positions is

$$D_i = k_i T - G_i.$$

It is divisible by H , hence by h_i . Moreover, Equation (4.13) gives

$$D_i \geq k_i(L - 1) + \kappa w F = \Omega(w^b).$$

Let $d_i = D_i/h_i$; for sufficiently large w , every $d_i > k_i$. Allocate d_i disjoint ordered groups of q_i wires to run the fixed h_i -call identity word for A_i . All these groups, for all types together, fit in the union of the all-identity sectors. Indeed that region has exactly $((1 - \theta)w)^b$ wires, whereas

$$\sum_i q_i d_i \leq \sum_i q_i k_i T = \theta w T < b\theta w^b < (1 - \theta)^b w^b. \quad (4.14)$$

We used $h_i \geq 1$ and $G_i \geq 0$ in the first inequality. A dummy group may cross the boundaries of the original identity sectors, since the prescribed action on their entire union is identity.

List the D_i idle positions for type i in chronological order, with any fixed order among positions in the same slot. Assign successive positions cyclically to its d_i dummy groups. Each group receives exactly h_i positions because $D_i = d_i h_i$. Two successive positions assigned to the same group are d_i places apart in this list. A single time contains at most $k_i < d_i$ positions, so these two positions have strictly increasing times. Assign the identity word's successive calls to these times. This argument permits arbitrary clustering of vacancies. We now have exactly k_i calls to every type i in every slot, on mutually disjoint tuples.

The exact word, including all monomials. The physical support of each active-sector program is fixed. Dummy groups have fixed supports disjoint from each other and from those sectors. Write any one of these programs chronologically as

$$O_{p,0}, A_{p,1}, O_{p,1}, \dots, A_{p,l_p}, O_{p,l_p},$$

where the $O_{p,j}$ are monomials on that program's support. Let its strictly increasing call times be $t_{p,1}, \dots, t_{p,l_p}$. Place $O_{p,0}$ in the gap immediately before $t_{p,1}$, and place $O_{p,j}$, $j \geq 1$, in the gap immediately after $t_{p,j}$. This is always before its next call, including when the times are consecutive.

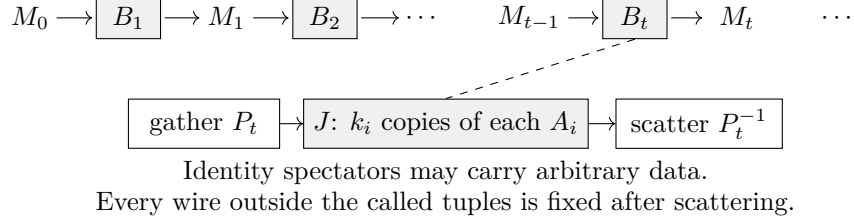


Figure 1: Chronological gaps preserve each local word. Each simultaneous call slot is one embedded J call with its gather and scatter permutations.

A program with no calls can place its single monomial in the initial gap. Operations in a gap on different program supports combine into one ambient monomial M_t , where gap t follows slot t , and gap zero precedes slot one. A sector monomial may permute all values on its support. Subsequent calls still use the fixed tuple indices prescribed by that same program, so this causes no change to the program's meaning.

Let B_t be the simultaneous calls in slot t . There are exactly k_i ordered tuples for each A_i . Match them bijectively to the corresponding active blocks of J . Their union has $\sum_i q_i k_i = \theta w$ wires. Complete this matching to an injection of all w coordinates of J by using $(1 - \theta)w$ further distinct wires as spectators for its identity blocks. There are enough: the ambient width is $w^b \geq w$. Extend the matching to a permutation P_t of all ambient wires. With the convention that P_t gathers into the canonical first w positions, one has exactly

$$B_t = P_t^{-1}(J \oplus I_{w^b - w})P_t. \quad (4.15)$$

The conjugate acts by the requested A_i on every called tuple and fixes every other physical wire pointwise. In particular a spectator can belong to an unfinished sector program or dummy word: both its value and its location are preserved by the complete gather–call–scatter operation. Figure 1 shows the chronological slots and the gather–call–scatter realization of each B_t .

Thus the matrix product

$$M_T B_T M_{T-1} \cdots M_1 B_1 M_0 \quad (4.16)$$

restricts to the prescribed program on every active sector and to identity on every dummy group and unused identity wire. It is exactly $J^{\otimes b}$. By Equation (4.15) it uses precisely $T < bw^{b-1}$ calls to J , with all remaining operations monomial, on precisely w^b wires. Since at least one A_i occurs in J , the matrix J is nonmonomial. This is a finite win, contrary to the assumption. $\square$

4.2 Separation, normalization, and compactness

For a nonempty finite comparison family on a nonempty finite type list, Theorem 4.2 says that the convex hull of its difference vectors is disjoint from the open positive orthant $(0, \infty)^m$. Finite-dimensional separation supplies a nonzero vector v and a real number c with

$$v \cdot \Delta^{(r)} \leq c \leq v \cdot y \quad \text{for all } r \text{ and all } y \in (0, \infty)^m;$$

see Boyd and Vandenberghe [7, Section 2.5.1, p. 46]. If a coordinate of v were negative, sending that coordinate of y to infinity would contradict the lower bound by c . Thus $v \geq 0$. The infimum of $v \cdot y$ on the positive orthant is zero, so $c \leq 0$. We have obtained nonnegative prices, not all zero, satisfying every comparison in the finite family. For an empty family any nonzero nonnegative vector suffices.

We next prevent the normalization from escaping along different types. Include the fixed shear U of Equation (4.6) in every finite type list. For each nonmonomial A of width q , fix the following two words once and for all. Gaussian elimination implements A by a finite positive number C_A of embedded U -calls and monomials. Indeed row permutations and nonzero row scalings are monomial, and every nonzero row addition is a diagonal conjugate of an ordered U -call. The comparison with one-factor target A imposes

$$p(A) \leq C_A p(U).$$

On width $2q$, apply Theorem 2.2 to $A \oplus I_q$. Its calls are embedded calls to A , and it generates $U \otimes I_q$ by some fixed word with $e_A > 0$ such calls and monomials. The two-factor target $U \otimes I_q$ has standard count q for U , giving

$$qp(U) \leq e_A p(A).$$

Consequently these two comparisons impose the fixed bounds

$$c_A p(U) \leq p(A) \leq C_A p(U), \quad c_A = q/e_A > 0. \quad (4.17)$$

They introduce no new nonmonomial type other than A and U . For $A = U$, take simply $c_U = C_U = 1$.

Augment any finite comparison family by these bounds for all its types. Separation gives a nonzero nonnegative solution. If its U -price were zero, every other coordinate would be zero by the upper bounds, which is impossible. Divide by its positive U -price to obtain $p(U) = 1$. Applying this argument just to the bounds for one type also proves that each fixed interval $[c_A, C_A]$ is nonempty.

The collection of all finite complex matrices is a set, as it is contained in $\bigcup_{n \geq 1} \mathbb{C}^{n^2}$. Form the product over all nonmonomial invertible matrices

$$\mathcal{P} = \prod_A [c_A, C_A].$$

Each factor is a nonempty compact interval, so this product is compact in the product topology by Tychonov's Theorem [19, Tag 08ZU, Theorem 5.14.4]. Each comparison specifies a closed subset of $\mathcal{P}$, since it is a linear inequality in finitely many coordinate projections. Any finite collection of these closed subsets has nonempty intersection: apply the preceding finite argument to its types, include U , and augment with their fixed bounds; all other coordinates can be assigned their lower endpoints. Compactness gives a point satisfying every comparison simultaneously. Call this preliminary function p_0 , and set $p_0(A) = 0$ on monomial matrices. It is strictly positive on each nonmonomial matrix, and $p_0(U) = 1$.

4.3 Algebraic laws and symmetry

We derive the laws from the comparisons before symmetrizing. A one-factor target AB , implemented by consecutive calls to B and A , gives $p_0(AB) \leq p_0(A) + p_0(B)$. This remains true if any of the matrices is monomial, since its call is then a monomial operation of price zero. Multiplication by monomials on either side cannot increase price. Applying the same observation with their inverses shows $p_0(LAR) = p_0(A)$.

The standard axis program for the one-factor target $A \otimes B$ gives

$$p_0(A \otimes B) \leq |B|p_0(A) + |A|p_0(B).$$

For the reverse inequality, use the two-factor target $A \otimes B$ and the word consisting of one call to that entire matrix. Its comparison gives the opposite inequality. Hence the tensor law is exact, including when a factor is an identity or any other monomial.

For direct sums the one-factor target $A \oplus B$, implemented by disjoint calls to its two blocks, gives $p_0(A \oplus B) \leq p_0(A) + p_0(B)$. The reverse inequality requires a separate construction. Put $a = |A|$, $e = |B|$, and

$$X = A^{\otimes a} \otimes B^{\otimes e}, \quad Q = |X|.$$

Consider the tensor target $X \otimes I_2$, with its factors listed as a copies of A , e copies of B , and I_2 . After a coordinate permutation this is two independent copies of X ; this statement is an identity of matrices and does not use a direct-sum price law. In the first copy run all A -axes and then all B -axes. In the second copy run all B -axes and then all A -axes. There are exactly $aQ/a = Q$ individual A -calls and $eQ/e = Q$ individual B -calls in each copy. Pair the successive Q first-stage A -calls of the first copy with the successive Q first-stage B -calls of the second copy. Each pair acts on disjoint tuples and is one embedded $A \oplus B$ -call, after gathering its coordinates. Pair the second stages in the same manner, reversing which copy supplies which type. Each copy retains its own required call order. We obtain a word with $2Q$ calls to $A \oplus B$, with only permutations added. The comparison for the displayed tensor target therefore gives

$$2Q(p_0(A) + p_0(B)) \leq 2Qp_0(A \oplus B).$$

The construction also applies when one or both factors are monomial: we may keep their individual block operations in the pairing, while their contribution to the comparison is zero. Thus direct-sum additivity is established without using it to produce the two copies.

Finally define

$$p(A) = \frac{1}{4}(p_0(A) + p_0(A^{-1}) + p_0(A^\top) + p_0(A^{-\top})). \quad (4.18)$$

Inversion and transposition commute and each has order two, so this function has both symmetries. Each of the four summands retains product subadditivity: transposition or inversion may reverse the order of AB , but the scalar upper bound is still $p_0(A) + p_0(B)$ with the corresponding transforms. They retain both exact laws because inverse and transpose act componentwise on tensor products and direct sums. They retain monomial invariance, and are positive exactly on nonmonomials, because these operations preserve the class of monomial matrices. Also U^{-1} is conjugate to U by $\text{diag}(1, -1)$, and $U^\top$ is conjugate to U by the two-coordinate swap. Thus every summand in Equation (4.18) has value one at U . This proves Theorem 4.1. All subsequent arguments use this symmetric p and its proved laws, without imposing additional comparison constraints on the average.

Corollary 4.3 (Shears, pairs, and embedded words). *Every nontrivial elementary coordinate shear has price one, including with any number of untouched coordinates. Every invertible 2×2 matrix has price at most two. If a word on w coordinates implements G , using nonmonomial calls $A_1, \dots, A_l$ and monomials, then*

$$p(G) \leq \sum_{j=1}^l p(A_j).$$

If the word instead implements $G \oplus I_s$, the same bound holds for $p(G)$, with no restriction on the initial values of the s coordinates restored by the word.

Proof. A shear adding a nonzero multiple of one coordinate into another is a monomial conjugate of $U \oplus I$, so its price is one by Equations (4.1), (4.4) and (4.6). For an invertible pair matrix, first permute rows if needed to make its upper-left entry nonzero. With entries $\alpha, \beta, \gamma, \delta$ and determinant $D \neq 0$, it factors as

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ \gamma/\alpha & 1 \end{pmatrix} \begin{pmatrix} \alpha & 0 \\ 0 & D/\alpha \end{pmatrix} \begin{pmatrix} 1 & \beta/\alpha \\ 0 & 1 \end{pmatrix}.$$

This uses at most two nontrivial shears and a monomial. Finally an embedded call to A is a permutation conjugate of $A \oplus I_{w-|A|}$, omitting the identity block when it has size zero, whose price is $p(A)$. Repeated product subadditivity gives the word bound. Identity padding has price zero by direct-sum additivity. The last assertion is an exact matrix identity on all input coordinates; it does not assume zero scratch values. $\square$

5 Triangular matrices and invertible corners

Continue under the no-finite-win assumption and fix the symmetric price of Theorem 4.1. Our goal is contraction to an invertible submatrix.

Theorem 5.1 (Invertible corners). *If an invertible matrix K has an invertible square submatrix M , then $p(M) \leq p(K)$.*

Independent row and column permutations move M to a corner without changing the price of K . To compare their prices, we will run the resulting block matrix along many tensor axes and for many time steps. The other coordinates persist as state between successive steps. Their initial values affect the output, so the main task is to remove that contribution at a cost small compared with the number of calls.

We first obtain price bounds for rectangular shears and scalar convolutions. We then prove that a block triangular matrix costs at least the sum of its diagonal blocks. These tools allow us to correct the initial state in a long cascade and isolate the repeated tensor powers of M that prove the theorem. The one-way price bound for a scalar circuit is a consequence of the corner theorem, derived at the end of the section; the preliminary convolution estimate uses circuits in both directions.

5.1 Rectangular shears and scalar convolution

For $G \in \text{Mat}_{a \times b}(\mathbb{C})$ put

$$s(G) = p \left(\begin{pmatrix} I_a & G \\ 0 & I_b \end{pmatrix} \right).$$

Its action on (y, x) is $(y + Gx, x)$. Empty blocks can be omitted.

Lemma 5.2 (A rectangular shear from a linear DAG). *Suppose a permitted linear DAG with l scalar gates computes Gx , with a designated output coordinates. Then*

$$s(G) \leq 8l + 2a. \tag{5.1}$$

If only a_0 rows of G are nonzero, a may be replaced by a_0 . The constant is independent of the dimensions and all circuit scalars.

Proof. A finite DAG has finite depth and finite maximum fan-out, counting repeated operand edges and designated output uses. Apply Theorem 3.1 to this DAG with source coordinates x , target coordinates y , and one additional coordinate per gate, with arbitrary initial values. It gives at most $8l + 2a$ elementary shears realizing $(x, y, z) \mapsto (x, y + Gx, z)$. Although depth and fan-out may vary with the DAG, the shear count is independent of both; the layer bound is not needed here. By Theorem 4.3 and identity padding, this proves Equation (5.1). If only a_0 rows are nonzero, apply the same replay to the submap of those rows, retaining all l gate coordinates, and identity-pad the omitted targets. This gives $8l + 2a_0$ instead; if $a_0 = 0$, the shear is identity. $\square$

For an invertible $m \times m$ matrix G , the exact identity

$$\begin{pmatrix} I & G \\ 0 & I \end{pmatrix} \begin{pmatrix} I & 0 \\ -G^{-1} & I \end{pmatrix} \begin{pmatrix} I & G \\ 0 & I \end{pmatrix} = \begin{pmatrix} 0 & G \\ -G^{-1} & 0 \end{pmatrix} \quad (5.2)$$

gives

$$2p(G) \leq 2s(G) + s(G^{-1}). \quad (5.3)$$

Indeed the right side of Equation (5.2) is monomial-equivalent to $G \oplus G^{-1}$, while exchanging the two coordinate groups and changing signs identifies the middle shear's price with $s(G^{-1})$. Consequently circuits for both G and G^{-1} , of sizes l and l' , give $p(G) = O(l + l' + m)$ with a universal constant.

For a scalar or matrix formal power series $F(z)$, let $\mathcal{T}_t(F)$ be multiplication by $F(z)$ modulo z^t , in the coefficient basis ordered with time first. Thus, if $F(z) = \sum_{j \geq 0} F_j z^j$, its (a, b) block is F_{a-b} for $0 \leq b \leq a < t$, and is zero otherwise. Coefficients with negative indices mean zero. Series multiplication gives

$$\mathcal{T}_t(FG) = \mathcal{T}_t(F)\mathcal{T}_t(G). \quad (5.4)$$

In particular, invertibility of F_0 implies invertibility of $\mathcal{T}_t(F)$, with inverse $\mathcal{T}_t(F^{-1})$.

Lemma 5.3 (Scalar convolution). *For every scalar formal series f with $f(0) \neq 0$ and every $t \geq 1$,*

$$p(\mathcal{T}_t(f)) = O(t \log(2t)),$$

with an absolute implied constant, independent of the coefficients of f and of their possible dependence on t .

Proof. Only the first t coefficients of f affect the matrix. Pad these coefficients and the input coefficient vector to a power-of-two length m with $2t \leq m < 4t$. A length- m Fourier transform, multiplication by the prechosen Fourier transform of the fixed kernel, and an inverse Fourier transform compute their ordinary convolution, since its degree is at most $2t - 2 < m$. Selecting the first t coefficients gives $\mathcal{T}_t(f)$. The binary Fourier recursion splits even and odd indices and combines each pair by one scalar multiplication and two additions or subtractions, so it and its inverse use $O(m \log(2m))$ permitted scalar gates. The frequency scalings and inverse normalization cost $O(m)$ gates, even when some kernel frequencies are zero. This is a DAG construction and does not require those intermediate scalings to be invertible.

Exactly the same construction applies to the first t coefficients of $1/f$, which exist because $f(0) \neq 0$. Apply Theorem 5.2 and Equation (5.3) to these two DAGs. Every kernel coefficient and Fourier constant is prechosen; there is no charge for generating its description, but its multiplication on variable data has been included in the count. $\square$

5.2 Triangular monotonicity by whole-block compression

Lemma 5.4 (Triangular monotonicity). *If $M \in \text{GL}_n(\mathbb{C})$, $D \in \text{GL}_r(\mathbb{C})$ and $C \in \text{Mat}_{n \times r}(\mathbb{C})$, then*

$$p\left(\begin{pmatrix} M & C \\ 0 & D \end{pmatrix}\right) \geq p(M) + p(D). \quad (5.5)$$

The same conclusion holds for the lower triangular orientation, and the price of any invertible block triangular matrix is at least the sum of the prices of its invertible diagonal blocks.

Proof. Write $K = \begin{pmatrix} M & C \\ 0 & D \end{pmatrix}$ and fix $k \geq 1$. The usual computation of $Y = D^{\otimes k}$ on $R = r^k$ bottom coordinates applies D along axes $1, \dots, k$, using $u = kr^{k-1}$ calls. Replace each call by a K call, supplying a fresh group of n data coordinates for its M side. Since these groups are used only once and do not affect the bottom coordinates, the resulting matrix is

$$\begin{pmatrix} I_u \otimes M & E \\ 0 & Y \end{pmatrix}, \quad p\left(\begin{pmatrix} I_u \otimes M & E \\ 0 & Y \end{pmatrix}\right) \leq up(K).$$

We will compress the u whole $n \times R$ row blocks of EY^{-1} . No independent change of its individual scalar rows is used.

Label a call by its axis j and fiber indices $\mathbf{i} = (i_\ell)_{\ell \neq j}$. For $1 \leq a \leq n$ and a bottom multi-index $\ell = (\ell_1, \dots, \ell_k)$, its row block $R_{j,\mathbf{i}}$ in EY^{-1} has entries

$$(R_{j,\mathbf{i}})_{a,\ell} = (CD^{-1})_{a,\ell_j} \prod_{h < j} \delta_{i_h, \ell_h} \prod_{h > j} (D^{-1})_{i_h, \ell_h}. \quad (5.6)$$

To see this, the call's contamination uses C at axis j , with D already applied on axes $h < j$ and identity on axes $h > j$. Multiplication by Y^{-1} cancels the earlier D factors, leaves CD^{-1} locally, and introduces D^{-1} on the later axes.

For fixed j , insertion of the fixed matrix CD^{-1} in the j th slot defines a linear map from the complementary tensor covector space to $\text{Mat}_{n \times R}(\mathbb{C})$. The complementary covectors in Equation (5.6) form a basis, because every D^{-1} is invertible. Coordinate covectors form another basis. The images of either basis therefore span the same image space, whether or not this linear map is injective. Images of coordinate covectors have at most nr nonzero entries. Taking their union over j gives a sparse spanning family for the span S of all actual row blocks. Select a basis $B_1, \dots, B_h$ from this family; then

$$h \leq \min(u, nR), \quad \text{nnz}(B_i) \leq nr.$$

Overlaps between axis spans and dependencies among rows of C can only decrease h .

The map $\Phi : \mathbb{C}^u \rightarrow S$ sending coefficients to their linear combination of the actual row blocks is surjective. Choose one lift of each B_i under Φ , and append a basis of $\ker \Phi$. These u vectors form a basis of $\mathbb{C}^u$: applying Φ to a relation first kills the coefficients of the B_i , and then independence in the kernel kills the others. Put these vectors as the rows of $V \in \text{GL}_u(\mathbb{C})$. Postapply $V \otimes I_n$ to the data groups. The data block becomes exactly

$$(V \otimes I_n)(I_u \otimes M) = V \otimes M,$$

while its contamination expressed in the final bottom variables is the stack $B_1, \dots, B_h, 0, \dots, 0$. Cancel it by at most hnr elementary shears from those bottom variables into the data groups. The final map is exactly $(V \otimes M) \oplus Y$.

The previously proved price laws now give

$$np(V) + up(M) + up(D) \leq up(K) + np(V) + hnr.$$

The identical $np(V)$ terms cancel, without any estimate on V . Since $h \leq nr^k$, division by $u = kr^{k-1}$ yields

$$p(M) + p(D) \leq p(K) + \frac{n^2 r^2}{k}.$$

Let k tend to infinity with K, n, r fixed. This proves Equation (5.5). If $r = 1$, the same construction has $R = 1$, $u = k$ and error at most n^2/k . If $S = 0$, there are no B_i and a kernel basis alone supplies

V ; no cancellation shears are needed. Empty diagonal blocks simply reduce to the other block. Exchanging the two groups proves the lower triangular case, and induction on the number of blocks proves the last assertion. $\square$

5.3 A time cascade and its initial-state correction

Fix an invertible cell

$$K = \begin{pmatrix} M & C \\ B & D \end{pmatrix}, \quad M \in \text{GL}_n(\mathbb{C}), \quad D \in \text{Mat}_{r \times r}(\mathbb{C}), \quad H(z) = M + zC(I_r - zD)^{-1}B. \quad (5.7)$$

No invertibility of D is required. The transfer matrix H is regular at zero and $H(0) = M$. For one fiber, with incoming state w_τ , input x_τ and output y_τ , the cell equations are

$$y_\tau = Mx_\tau + Cw_\tau, \quad w_{\tau+1} = Bx_\tau + Dw_\tau.$$

Writing series in time and retaining coefficients below t gives

$$w(z) = w_0 + zBx(z) + zDw(z), \quad y(z) = H(z)x(z) + C(I_r - zD)^{-1}w_0 \pmod{z^t}.$$

These are identities between finite coefficient maps, rather than operations that a circuit is asked to perform implicitly.

For $k \geq 1$ put $N = n^k$. At each tick, apply the cell along each of the k spatial tensor axes in order, using n^{k-1} disjoint fibers per axis. Axis j has its own persistent state space

$$W_j = (\mathbb{C}^n)^{\otimes(j-1)} \otimes \mathbb{C}^r \otimes (\mathbb{C}^n)^{\otimes(k-j)}.$$

States of different axes are distinct, and each W_j is carried from one tick to the next. The t data blocks are fresh inputs, one per tick. The entire computation is a word in invertible K calls on $tN + k rn^{k-1}$ coordinates; denote its matrix by $\mathcal{K}_{k,t}$. Figure 2 shows two successive ticks and the distinct state spaces carried by the axes from one tick to the next.

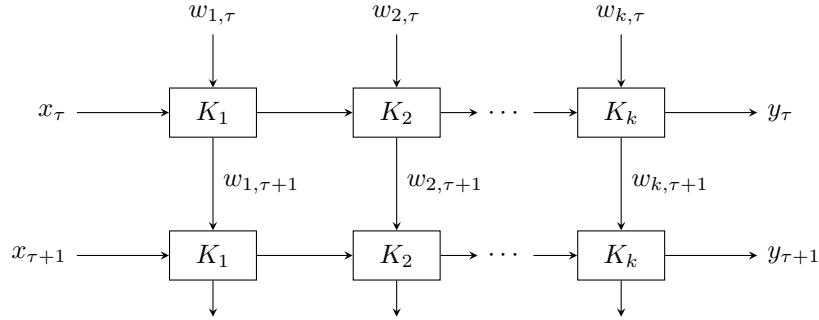


Figure 2: Two ticks of the cascade. Each K_j represents n^{k-1} disjoint calls along axis j . Horizontal arrows carry N data coordinates; the separate vertical arrows carry each axis's rn^{k-1} persistent state coordinates.

With data coordinates listed before all state coordinates, the top-left block of $\mathcal{K}_{k,t}$ is

$$A_{k,t} = \mathcal{T}_t(H_k), \quad H_k(z) = H(z)^{\otimes k}.$$

Indeed the direct series response composes the lifts of H on all axes. The top-right block E_t consists of the first t coefficients of $E(z) = [E_1(z) \cdots E_k(z)]$, where

$$E_j(z) = I_n^{\otimes(j-1)} \otimes (C(I_r - zD)^{-1}) \otimes H(z)^{\otimes(k-j)}. \quad (5.8)$$

An initial state at axis j passes through that axis's state-to-data map and then through the later axes; the earlier axes do not act on this contribution. This proves Equation (5.8) as well as the direct response, including the order of all factors.

Lemma 5.5 (Uniform cascade estimate). *For each fixed cell K as in Equation (5.7), there is a constant C_K such that, for every $k, t \geq 1$ and $N = n^k$,*

$$p(\mathcal{T}_t(H^{\otimes k})) \leq t k n^{k-1} p(K) + C_K (N t \log(2t) + (k+1)^3 N). \quad (5.9)$$

The constant is independent of t , k , and any scalars used for coefficient interpolation.

Proof. If $r = 0$, the state is absent and $H = M = K$ is constant. The exact tensor and direct-sum laws already give the assertion. Suppose $r \geq 1$ and introduce polynomial matrices and scalar polynomials

$$\begin{aligned} a(z) &= \det(I_r - zD), & R(z) &= \text{adj}(I_r - zD), \\ F(z) &= a(z)M + zCR(z)B, & b(z) &= \det F(z), & J(z) &= \text{adj } F(z). \end{aligned}$$

Then $H = F/a$, $H^{-1} = aJ/b$, $a(0) = 1$, and $b(0) = \det M \neq 0$. We have

$$\deg a \leq r, \quad \deg R \leq r - 1, \quad \deg F \leq r, \quad \deg b \leq nr, \quad \deg J \leq (n - 1)r.$$

For $n = 1$ the adjugate J is the constant 1, as required by these formulas. Cancelling the later-axis responses in Equation (5.8) gives

$$H_k^{-1} E_j = (H^{-1})^{\otimes(j-1)} \otimes (JCR/b) \otimes I_n^{\otimes(k-j)}.$$

Thus $q_k = b^k$ clears the entire matrix $H_k^{-1} E$. Its j th cleared block is the following explicit polynomial map $W_j \rightarrow \mathbb{C}^N$:

$$Q_{k,j}(z) = a(z)^{j-1} b(z)^{k-j} \left(J(z)^{\otimes(j-1)} \otimes (J(z)CR(z)) \otimes I_n^{\otimes(k-j)} \right). \quad (5.10)$$

Its degree is at most

$$(j-1)r + (k-j)nr + (j-1)(n-1)r + (nr-1) = knr - 1.$$

The scalar q_k has degree at most knr and has nonzero constant term. The identities also hold when any of the displayed matrix polynomials vanish.

Here is an explicit DAG bound for the nonzero time coordinates of $Q_k(z)w = \sum_{j=1}^k Q_{k,j}(z)w_j$. Put $m = knr$, and choose m distinct complex evaluation points. Because Equation (5.10) is polynomial, no avoidance of poles is needed. At one evaluation point, a rectangular application of JCR on its r -axis costs at most $2rN$ scalar gates. Applying J on the preceding $j-1$ axes costs at most $2n(j-1)N$ gates, by doing the fixed $n \times n$ map on all fibers of each axis. The scalar factor costs at most N gates. Adding the k contributions costs at most $(k-1)N$ more. One evaluation therefore uses at most

$$(nk(k-1) + 2(r+1)k-1)N = O_K(k^2 N)$$

gates. All local entries and scalar powers here are prechosen constants. Applying the inverse scalar Vandermonde matrix to the m evaluations, separately on each of the N coordinates, recovers all m coefficient vectors using at most $2m^2 N$ further scalar gates. The total is $O_K((k+1)^3 N)$. For any run length t , keep only the first $\min(t, m)$ coefficient vectors; all other rows of the truncated response are zero. This bound is consequently uniform in t .

Set $P = \mathcal{T}_t(1/q_k)$ and $A' = A_{k,t}(P \otimes I_N)$. In formal series modulo z^t , $A'^{-1}E_t$ is precisely the coefficient map of $q_k H_k^{-1}E = Q_k$ just constructed. Hence Theorem 5.2, with zero output rows omitted, bounds the price of the shear with block $-A'^{-1}E_t$ by $O_K((k+1)^3 N)$. This uses only the rectangular shear law, not a price bound for an arbitrary invertible circuit output.

Now form the exact product

$$\mathcal{K}_{k,t} \begin{pmatrix} P \otimes I_N & 0 \\ 0 & I \end{pmatrix} \begin{pmatrix} I & -A'^{-1}E_t \\ 0 & I \end{pmatrix}. \quad (5.11)$$

Its top row of blocks is $(A', 0)$. Each factor is invertible, and A' is invertible because $H(0) = M$ and $q_k(0) \neq 0$. Therefore the other diagonal block of this lower triangular product is also invertible. By Theorem 5.4 its price is at least $p(A')$. There are tkn^{k-1} cell calls in $\mathcal{K}_{k,t}$, so

$$p(A') \leq tkn^{k-1}p(K) + Np(P) + O_K((k+1)^3 N).$$

Here the tensor and direct-sum laws price the scalar preprocessing by $Np(P)$, including its identity on the state. Finally $A_{k,t} = A'(P^{-1} \otimes I_N)$ and inverse symmetry give

$$p(A_{k,t}) \leq tkn^{k-1}p(K) + 2Np(P) + O_K((k+1)^3 N).$$

Apply Theorem 5.3 to P . Its bound is uniform even though q_k varies with k , proving Equation (5.9). $\square$

5.4 Proof of the corner law and its circuit consequence

Proof of Theorem 5.1. Independent row and column permutations move M to the top-left corner and preserve $p(K)$. Write the resulting matrix as in Equation (5.7). If the complementary width is zero the claim is equality; otherwise apply Theorem 5.5. The matrix $A_{k,t} = \mathcal{T}_t(H^{\otimes k})$ is lower triangular in time with t identical diagonal blocks $M^{\otimes k}$. Thus Theorem 5.4 and the tensor law imply

$$p(A_{k,t}) \geq tp(M^{\otimes k}) = tkn^{k-1}p(M).$$

Fix K, n, r first and use the bound of Equation (5.9). For example, choose $t = t(k) = 2^{\lceil 8 \log_2(k+1) \rceil}$, so $(k+1)^8 \leq t < 2(k+1)^8$. Division by $tkn^{k-1} = tkN/n$ gives

$$p(M) \leq p(K) + C_K n \left(\frac{\log(2t)}{k} + \frac{(k+1)^3}{tk} \right).$$

Both errors tend to zero. The matrices M and K in this real inequality are fixed, so it proves the exact inequality $p(M) \leq p(K)$. No continuity assertion about the price, or limiting circuit for a matrix, has been used. A zero-size submatrix, if allowed, can simply be omitted and assigned price zero. $\square$

Corollary 5.6 (A universal one-way DAG price bound). *If $G \in \text{GL}_m(\mathbb{C})$ has a permitted linear DAG with l scalar gates, then*

$$p(G) \leq s(G) \leq 8l + 2m.$$

The bound holds with the same constants for all dimensions and all matrix families, without requiring a DAG for G^{-1} .

Proof. G is an invertible square submatrix of $\begin{pmatrix} I_m & G \\ 0 & I_m \end{pmatrix}$, so Theorem 5.1 gives its first inequality, and Theorem 5.2 gives the second. Although the proof of the corner law fixed each individual matrix while taking a limit, the resulting inequality is exact for every finite matrix. The numerical constants here come entirely from the dirty scratch simulation. $\square$

6 Changing the coefficient boundary and specializing prices

The corner law compares the price of an invertible matrix with that of an invertible submatrix. We now consider a different operation on the cell from the preceding section. Write

$$K = \begin{pmatrix} M & C \\ B & D \end{pmatrix} \in \mathrm{GL}_{n+r}(\mathbb{C}), \quad K(x, w) = (y, w'), \quad M \in \mathrm{GL}_n(\mathbb{C}).$$

Closing its state connection at a scalar s means imposing $w = sw'$. When $I_r - sD$ is invertible, the state equation gives

$$w = s(I_r - sD)^{-1}Bx, \quad y = H(s)x, \quad H(z) = M + zC(I_r - zD)^{-1}B.$$

Our first objective is to prove $p(H(s)) \leq p(K)$ whenever $H(s)$ is invertible. We will then prove that a uniform price bound for a rational matrix at generic scalar values remains valid at every regular invertible value. Closing the connection serves only to define the finite matrix $H(s)$; every price inequality concerns ordinary invertible matrices.

Both arguments use a common comparison between coefficient boundary conditions. The data-to-data block of the preceding cascade is multiplication by a matrix series modulo z^t . We compare this with multiplication modulo a different scalar polynomial of degree t . For a k th tensor power, a scalar change of coordinates confines the discrepancy to $O(k)$ time indices. We must also control the large spatial matrix on those indices: an explicit common family of inexpensive generators supplies this second bound. The resulting estimate is uniform over the scalar modulus, which can therefore be chosen separately at each tensor order.

We retain the time coefficient convention for $\mathcal{T}_t$: for a matrix series $F(z)$ regular at zero, $\mathcal{T}_t(F)$ sends the coefficients of $x(z)$, of degree less than t , to those of $F(z)x(z)$ modulo z^t , with the time index outermost. All prices in this section satisfy the laws of Theorem 4.1.

6.1 A uniform comparison for scalar moduli

Lemma 6.1 (Coefficient-boundary comparison). *Fix $G(z) \in \mathrm{Mat}_n(\mathbb{C}[z])$ with $G(0)$ invertible. There is a constant C_G with the following property. Let $k \geq 1$, put $N = n^k$ and $d = k \deg G$, and let $t > d$ be an integer. Suppose $\ell(z)$ is monic of degree t , $\ell(0) \neq 0$, and multiplication by $G(z)^{\otimes k}$ on $(\mathbb{C}[z]/(\ell))^N$ is invertible. Denote its matrix in the coefficient basis by W , and put*

$$A_G = \mathcal{T}_t(G(z)^{\otimes k}).$$

Then

$$|p(W) - p(A_G)| \leq C_G N (t \log(2t) + (k+1)^6). \quad (6.1)$$

The constant is independent of k, t, ℓ and of the magnitudes of the coefficients of ℓ . In particular, for

$$t = t(k) = 2^{\lceil 8 \log_2(k+1) \rceil}, \quad (k+1)^8 \leq t < 2(k+1)^8, \quad (6.2)$$

the difference is $o(tkN)$, uniformly over every admissible modulus of that degree.

Proof. Choose once and for all polynomial $J(z)$ and scalar polynomial $u(z)$ such that

$$G(z)^{-1} = J(z)/u(z), \quad u(0) \neq 0;$$

for example, take $J = \mathrm{adj} G$ and $u = \det G$. Write

$$G_k(z) = G(z)^{\otimes k} = \sum_{j=0}^d G_j z^j, \quad J_k(z) = J(z)^{\otimes k} = \sum_{i=0}^e J_i z^i, \quad e = k \deg J.$$

These are degree bounds with zero coefficient matrices permitted. The subscript on G_j or J_i in this proof denotes a coefficient of the tensor power, rather than of the original fixed matrix. Both d and e are bounded by a fixed multiple of k . If $d = 0$, multiplication is coefficientwise, so $W = A_G$ and the result holds. Assume $d > 0$ and set $T = t - d$.

Let $L = W - A_G$. An input supported at times less than T has product with G_k of degree less than t , so L has zero columns at those times. Scalar polynomial division also shows that, for every input $x(z)$ of degree less than t ,

$$Lx = \ell(z)v_x(z) \pmod{z^t}, \quad \deg v_x < d. \quad (6.3)$$

Here v_x is minus the quotient of $G_k x$ by ℓ , with scalar division applied to every coordinate. The degree bound follows from $\deg(G_k x) < t + d$.

The invertible relative map and the scalar prefix map are

$$R = A_G^{-1}W = I + S, \quad S = A_G^{-1}L, \quad P = \mathcal{T}_T(u^k/\ell) \oplus I_d.$$

The series in P is regular and nonzero at zero. Since S has zero prefix columns and P^{-1} is identity on the suffix of length d ,

$$S(P^{-1} \otimes I_N) = S, \quad \hat{R} = (P \otimes I_N)R(P^{-1} \otimes I_N) = I + (P \otimes I_N)S. \quad (6.4)$$

This identity uses the entire prefix–suffix decomposition; a general scalar change of basis on all t times would not have this property.

On the prefix, the output of the correction in Equation (6.4) is, modulo z^T ,

$$\frac{u^k}{\ell} \frac{J_k}{u^k} \ell v_x = J_k v_x.$$

It therefore vanishes at prefix times greater than or equal to $e + d$. Define the disjoint sets

$$\mathcal{E} = \{0, \dots, \min(T, e + d) - 1\}, \quad \mathcal{B} = \{T, \dots, t - 1\}, \quad \mathcal{U} = \mathcal{E} \cup \mathcal{B}.$$

The correction has columns only in $\mathcal{B}$ and rows only in $\mathcal{U}$, where $|\mathcal{U}| \leq e + 2d = O(k)$. Thus both blocks coupling $\mathcal{U}$ to its complement are zero in the correction. A simultaneous permutation of rows and columns gives

$$\hat{R} \simeq R_{\mathcal{U}} \oplus I_{(t-|\mathcal{U}|)N}, \quad R_{\mathcal{U}} \in \text{GL}_{|\mathcal{U}|N}(\mathbb{C}). \quad (6.5)$$

Invertibility of the active block follows from invertibility of $\hat{R}$. If the coarse early and suffix ranges would overlap, the minimum in $\mathcal{E}$ truncates the former: the middle set is then empty and $t \leq e + 2d$. The bound and the direct-sum conclusion still hold. Figure 3 displays the support of the correction $\hat{R} - I$ when the middle set is nonempty.

The small number of active times alone would not bound the cost of their $N \times N$ spatial blocks. We next give an indexed formula that controls those blocks uniformly. For each integer $m \geq 0$ that occurs below, divide the scalar monomial z^m by ℓ and write

$$z^m = \ell(z)q_m(z) + r_m(z), \quad \deg r_m < t, \quad q_{s,m} = [z^s]q_m(z).$$

For $m = b + j$ with $0 \leq b < t$ and $0 \leq j \leq d$, we have $\deg q_m < d$. Set $q_{s,m} = 0$ when $s < 0$ or $s \geq d$; also $q_m = 0$ when $m < t$. Let

$$\eta_h = [z^h](\ell/u^k), \quad \eta_h = 0 \quad (h < 0).$$

	early $\mathcal{E}$	middle	tail $\mathcal{B}$
early $\mathcal{E}$	0	0	*
middle	0	0	0
tail $\mathcal{B}$	0	0	*

support of $\widehat{R} - I$

Figure 3: After prefix conjugation, only the shaded blocks can be nonzero in the correction. Restoring the identity leaves the middle coordinates as an identity direct summand. The picture shows the case with a nonempty middle; the definition of $\mathcal{E}$ also covers an empty middle.

All these scalars may depend on k, t, ℓ . For $0 \leq a, b < t$, define

$$\gamma_{abij} = \begin{cases} -q_{a-i, b+j}, & a < T, \\ -\sum_{s=0}^{d-1} \eta_{a-i-s} q_{s, b+j}, & a \geq T. \end{cases} \quad (6.6)$$

Then the exact block formula is

$$\widehat{R}_{ab} = \delta_{ab} I_N + \sum_{i=0}^e \sum_{j=0}^d \gamma_{abij} J_i G_j. \quad (6.7)$$

Indeed, on an input $z^b x_b$, scalar division gives $v_x = -\sum_j q_{b+j} G_j x_b$ in Equation (6.3). The prefix correction is $J_k v_x$, which gives the first case of Equation (6.6). The suffix is unchanged by the left prefix map, so its correction is $(\ell/u^k) J_k v_x$ modulo z^t , which gives the second case. Right conjugation has already disappeared by Equation (6.4). Every spatial product is ordered as $J_i G_j$: no commutation of coefficient matrices is used. Reduction and scalar convolution introduce only the displayed scalar weights, irrespective of the length of the underlying coefficient lists.

Here is a gate count for applying the active block. Applying a fixed $n \times n$ matrix on each of the N/n fibers of one tensor axis uses at most $2nN$ scalar gates, by direct multiplication and addition. Consequently $G(\xi)^{\otimes k}$ applies in at most $2nkN$ gates for any $\xi \in \mathbb{C}$, including points where $G(\xi)$ is singular. At $d+1$ distinct points, interpolation expresses any chosen coefficient G_j as a scalar linear combination of these evaluation matrices. Running the $d+1$ evaluation circuits on the same input and combining their outputs uses at most

$$2nkN(d+1) + 2(d+1)N = O_G((k+1)^2 N)$$

gates. The same argument with $e+1$ points applies to J_i , so the ordered product $J_i G_j$ has a circuit of this order by composition. No inverse of an evaluation matrix is needed; only the scalar Vandermonde interpolation matrix is inverted in choosing constants.

There are $O_G((k+1)^2)$ active ordered pairs of time indices and $(e+1)(d+1) = O_G((k+1)^2)$ spatial generators in Equation (6.7). For each active block entry, apply each required generator to its input vector, multiply its output by γ_{abij} , and add the resulting vectors. The generator applications cost $O_G((k+1)^6 N)$ gates in total. The scalar weights and output additions cost only $O_G((k+1)^4 N)$ additional gates, and the identity contribution uses the already available input values.

The output count is $|\mathcal{U}|N = O_G((k+1)N)$. Thus the universal one-way DAG bound, Theorem 5.6, and Equation (6.5) give

$$p(\widehat{R}) = p(R_{\mathcal{U}}) \leq C_G(k+1)^6 N. \quad (6.8)$$

The potentially complicated coefficients in Equation (6.6) are prechosen circuit scalars. Every use of such a scalar has been charged; the model does not charge for producing its description. This is precisely why the gate bound is independent of t and ℓ .

By Theorem 5.3, direct-sum additivity and the tensor law,

$$p(P) = O(T \log(2T)) = O(t \log(2t)), \quad p(P \otimes I_N) = Np(P).$$

Undoing the conjugation and using inverse symmetry gives

$$p(R) = p(R^{-1}) \leq 2Np(P) + C_G(k+1)^6 N.$$

Finally $W = A_G R$ and $A_G = W R^{-1}$ imply $|p(W) - p(A_G)| \leq p(R)$, proving Equation (6.1). Dividing that bound by tkN gives

$$\frac{|p(W) - p(A_G)|}{tkN} \leq C_G \left(\frac{\log(2t)}{k} + \frac{(k+1)^6}{tk} \right). \quad (6.9)$$

For the choice of t in Equation (6.2), both terms tend to zero. Also $t > d$ eventually, since G was fixed before k grows. Every estimate holds uniformly for all admissible ℓ at that k, t . $\square$

6.2 Closing a state connection

Theorem 6.2 (Feedback law). *Let*

$$K = \begin{pmatrix} M & C \\ B & D \end{pmatrix} \in \text{GL}_{n+r}(\mathbb{C}), \quad M \in \text{GL}_n(\mathbb{C}), \quad H(z) = M + zC(I_r - zD)^{-1}B.$$

If $s_0 \in \mathbb{C}$ satisfies

$$\det(I_r - s_0 D) \neq 0, \quad \det H(s_0) \neq 0,$$

then

$$p(H(s_0)) \leq p(K). \quad (6.10)$$

No invertibility of D is required.

Proof. If there is no state block, $H = M = K$. With a state block present, $s_0 = 0$ is the invertible corner law, Theorem 5.1. We prove the claim for $s_0 \neq 0$ by fixing K, s_0 throughout the following estimates. Let

$$q(z) = \det(I_r - zD), \quad G(z) = q(z)M + zC \operatorname{adj}(I_r - zD)B = q(z)H(z).$$

Then G is polynomial, $q(0) = 1$, $q(s_0) \neq 0$ and $G(0) = M$ is invertible. Put $N = n^k$ and choose $t = t(k)$ as in Equation (6.2). For sufficiently large k , this choice meets the degree condition in Theorem 6.1.

Write $A_H = \mathcal{T}_t(H(z)^{\otimes k})$. The cascade estimate in Equation (5.9) gives, with constants depending only on the fixed cell,

$$p(A_H) \leq t k n^{k-1} p(K) + C_K N (t \log(2t) + (k+1)^3). \quad (6.11)$$

Moreover,

$$A_G = (\mathcal{T}_t(q^k) \otimes I_N) A_H,$$

and the scalar factor is invertible. The scalar-convolution bound and the product law in both directions show that

$$|p(A_G) - p(A_H)| \leq CNt \log(2t). \quad (6.12)$$

Its constant is independent of the degree or coefficients of q^k .

Apply the coefficient-boundary comparison with $\ell(z) = (z - s_0)^t$. To verify its remaining invertibility hypothesis, translate a polynomial to its coefficients at s_0 . The scalar translation matrix V_{s_0} has entries

$$(V_{s_0})_{ij} = \begin{cases} \binom{j}{i} s_0^{j-i}, & j \geq i, \\ 0, & j < i, \end{cases} \quad 0 \leq i, j < t.$$

It is invertible with inverse V_{-s_0} . In the translated coordinates, multiplication modulo $(z - s_0)^t$ becomes multiplication by $G(s_0 + w)^{\otimes k}$ modulo w^t . Hence

$$(V_{s_0} \otimes I_N) W (V_{s_0}^{-1} \otimes I_N) = \mathcal{T}_t(G(s_0 + w)^{\otimes k}). \quad (6.13)$$

The right-hand side is lower triangular in time, with repeated invertible diagonal block $G(s_0)^{\otimes k}$. Thus W is invertible, as required.

The price of translation is also small. If $D_! = \text{diag}(0!, 1!, \dots, (t-1)!)$ and U_{s_0} is the upper triangular Toeplitz matrix with entry $s_0^{j-i}/(j-i)!$ at $j \geq i$, then

$$V_{s_0} = D_!^{-1} U_{s_0} D_!.$$

Monomial invariance, transpose symmetry and Theorem 5.3 imply

$$p(V_{s_0}) = O(t \log(2t)). \quad (6.14)$$

Applying triangular monotonicity, Theorem 5.4, to Equation (6.13), then undoing its two scalar changes of basis, gives

$$\begin{aligned} p(W) &\geq t p(G(s_0)^{\otimes k}) - 2N p(V_{s_0}) \\ &= t k n^{k-1} p(H(s_0)) - O(Nt \log(2t)). \end{aligned} \quad (6.15)$$

For the equality, use the tensor law and the nonzero scalar relation $G(s_0) = q(s_0)H(s_0)$, which preserves price.

Combining Equations (6.1), (6.11), (6.12) and (6.15) yields

$$t k n^{k-1} (p(H(s_0)) - p(K)) \leq C_{K, s_0} N (t \log(2t) + (k+1)^6).$$

Divide by $t k n^{k-1} = t k N / n$ and let k tend to infinity with the single fixed choice of t in Equation (6.2). The error vanishes by Equation (6.9), proving Equation (6.10). $\square$

6.3 Specialization at any regular invertible value

Theorem 6.3 (Algebraic specialization). *Let $X(z) \in \text{Mat}_n(\mathbb{C}(z))$ be regular at $z_0 \in \mathbb{C}$, and assume $X(z_0)$ is invertible. Suppose there are a finite set $E \subset \mathbb{C}$ and a real constant C_0 such that, for every $s \notin E$, the matrix $X(s)$ is defined and invertible and*

$$p(X(s)) \leq C_0.$$

Then $p(X(z_0)) \leq C_0$.

Proof. Replace $X(z)$ by $X(z_0 + z)$ and E by $E - z_0$, so it suffices to treat $z_0 = 0$. By regularity, the entries have polynomial denominator representations nonzero at zero. Their product gives a scalar polynomial q with $q(0) \neq 0$ for which $G = qX$ is polynomial. In particular $G(0)$ is invertible. Enlarge the finite exceptional set to

$$E' = E \cup \{s : q(s) = 0\}.$$

At every $s \notin E'$, the matrix $G(s)$ is invertible and $p(G(s)) = p(X(s)) \leq C_0$.

Fix X, q, E' before letting k grow, put $N = n^k$, and again choose the single run length specified in Equation (6.2). For each such finite t , let ζ_t be the specified primitive t th root of unity. Choose

$$a = a_k \notin \{0\} \cup \bigcup_{h=0}^{t-1} \zeta_t^{-h} E'. \quad (6.16)$$

Only finitely many complex numbers are forbidden, so this choice exists for every k . No fixed dilation is asserted to work for all k , and no bound on a_k is needed. The polynomial

$$\ell(z) = z^t - a^t$$

has nonzero constant term and distinct roots $s_h = a\zeta_t^h$, all outside E' . Let W be multiplication by G_k modulo this polynomial. Scalar evaluation at the roots has matrix

$$V_a = (s_h^j)_{0 \leq h, j < t} = F_t \operatorname{diag}(1, a, \dots, a^{t-1}).$$

The roots are distinct, so this matrix is invertible, and evaluation gives the exact conjugacy

$$(V_a \otimes I_N)W(V_a^{-1} \otimes I_N) = \bigoplus_{h=0}^{t-1} G(s_h)^{\otimes k}. \quad (6.17)$$

This also proves that W is invertible.

Since t is a power of two, the binary FFT computes F_t with $O(t \log(2t))$ scalar gates. The universal DAG-price bound and monomial invariance therefore give

$$p(V_a) = p(F_t) = O(t \log(2t)),$$

uniformly in a . The direct-sum and tensor laws applied to Equation (6.17) imply

$$\begin{aligned} p(W) &\leq \sum_{h=0}^{t-1} p(G(s_h)^{\otimes k}) + 2Np(V_a) \\ &\leq tkn^{k-1}C_0 + O(Nt \log(2t)). \end{aligned} \quad (6.18)$$

On the other hand, $A_G = \mathcal{T}_t(G_k)$ is lower triangular with diagonal $G(0)^{\otimes k}$ repeated t times. Thus

$$p(A_G) \geq tkn^{k-1}p(G(0)) = tkn^{k-1}p(X(0)). \quad (6.19)$$

For sufficiently large k the degree condition of Theorem 6.1 holds. Its uniform estimate applies to the modulus just chosen, even though $a = a_k$ can vary. Combining it with Equations (6.18) and (6.19) and dividing by tkN/n proves

$$p(X(0)) \leq C_0 + C_X \left(\frac{\log(2t)}{k} + \frac{(k+1)^6}{tk} \right).$$

Letting k tend to infinity proves the assertion. $\square$

The limits in this section concern real inequalities between prices of exact finite matrices. Regularity of a rational matrix supplies its value at the specialization point, but no continuity of p is assumed or inferred. In particular, the exceptional set in a later application may contain the specialization point itself.

7 Pivoting and the price of a dense pair

Throughout this Section, p is the price function of Theorem 4.1, under the assumption that no finite win exists. By Theorem 4.3, identity padding does not change price, every nontrivial elementary coordinate shear has price one, and every invertible two-coordinate matrix has price at most two.

7.1 A general pivot inequality

Lemma 7.1 (Pivot inequality). *Let*

$$K = \begin{pmatrix} A & C \\ B & D \end{pmatrix} \in \mathrm{GL}_{n+r}(\mathbb{C}), \quad A \in \mathrm{GL}_n(\mathbb{C}), \quad D \in \mathrm{GL}_r(\mathbb{C}).$$

Write the action of K as $(x, z_{\mathrm{in}}) \mapsto (x', z_{\mathrm{out}})$. The map $(x, z_{\mathrm{out}}) \mapsto (x', z_{\mathrm{in}})$ is the invertible matrix

$$\widetilde{K} = \begin{pmatrix} A - CD^{-1}B & CD^{-1} \\ -D^{-1}B & D^{-1} \end{pmatrix}, \quad p(\widetilde{K}) \leq p(K) + 4r. \quad (7.1)$$

Proof. Solving $z_{\mathrm{out}} = Bx + Dz_{\mathrm{in}}$ gives the displayed formula. Its determinant is $\det A / \det D$, as follows by taking the Schur complement of D^{-1} , so it is invertible. Both pivot hypotheses are relevant: D makes the solve possible, whereas A makes its resulting transition invertible.

We realize this solve as a feedback operation to which Theorem 6.2 applies. Use data (x, y) of widths n, r and state z of width r . First apply, separately on the r pairs (y_i, z_i) , the matrix

$$P_\beta = \begin{pmatrix} \beta & 1 \\ 1 & 1 \end{pmatrix}, \quad (\widehat{y}, z_{\mathrm{in}}) = (\beta y + z, y + z).$$

Next apply K on (x, z_{in}) , leaving $\widehat{y}$ fixed. Finally apply

$$Q_\beta = \begin{pmatrix} 1 & 1 - \beta \\ 1 & -\beta \end{pmatrix} \quad \text{on each pair } (\widehat{y}_i, z_{\mathrm{out}, i}),$$

with outputs (y', z') . The full cell, denoted $\mathcal{K}_\beta$, sends (x, y, z) to (x', y', z') . Since $\det P_\beta = \beta - 1$ and $\det Q_\beta = -1$, it is invertible whenever $\beta \neq 1$, and

$$p(\mathcal{K}_\beta) \leq p(K) + 4r. \quad (7.2)$$

The blocks of this cell with respect to its data/state split are

$$\mathcal{K}_\beta = \left(\begin{array}{cc|c} A & C & C \\ (1 - \beta)B & \beta I_r + (1 - \beta)D & I_r + (1 - \beta)D \\ -\beta B & \beta(I_r - D) & I_r - \beta D \end{array} \right). \quad (7.3)$$

Its direct data block

$$M_\beta = \begin{pmatrix} A & C \\ (1 - \beta)B & \beta I_r + (1 - \beta)D \end{pmatrix}$$

has polynomial determinant and satisfies $M_0 = K$. Thus the precise genericity requirement is

$$\beta \notin \{0, 1\} \cup \{b \in \mathbb{C} : \det M_b = 0\}. \quad (7.4)$$

This excludes only finitely many numbers and hence allows a choice of β . At feedback parameter one, the feedback denominator is $I_r - (I_r - \beta D) = \beta D$, which is invertible.

Indeed, closing the state by $z = z'$ in $z' = \beta y + z - \beta z_{\text{out}}$ forces $z_{\text{out}} = y$. Consequently

$$y' = \beta y + z + (1 - \beta)z_{\text{out}} = y + z = z_{\text{in}}.$$

The feedback output map on (x, y) is therefore exactly $\widetilde{K}$, already proved invertible. The full cell, its direct block, its feedback denominator, and its output all meet the hypotheses of Theorem 6.2. That Theorem and Equation (7.2) prove the inequality. $\square$

An empty data block gives $\widetilde{K} = D^{-1}$ and the inequality directly from inverse symmetry. An empty pivot block makes no change. We use these conventions if either block has width zero.

7.2 One shared coordinate and exact scalar pivots

Lemma 7.2 (One-coordinate intersection). *Suppose a call to $Y_1 \in \text{GL}_a(\mathbb{C})$ is followed by a call to $Y_2 \in \text{GL}_b(\mathbb{C})$, and their ordered coordinate sets intersect in exactly one coordinate. If P is their product on the union of these sets, then*

$$p(P) = p(Y_1) + p(Y_2).$$

The same equality holds with any additional identity coordinates.

Proof. Let the shared coordinate occupy port i_0 of Y_1 and port j_0 of Y_2 . Index the tensor grid by $(i, j) \in \{1, \dots, a\} \times \{1, \dots, b\}$. The usual word for $Y_1 \otimes Y_2$ first applies Y_1 along each fiber with fixed j , and then Y_2 along each fiber with fixed i . Calls within either stage have disjoint coordinate sets, so put the $j = j_0$ call last in the first stage and the $i = i_0$ call first in the second stage. These two consecutive calls have the prescribed port identification: send port i of Y_1 to (i, j_0) and port j of Y_2 to (i_0, j) . Figure 4 shows the selected fibers of this consecutive call pair and their single shared coordinate.

Their union has $a + b - 1$ coordinates. On all other grid coordinates their product is identity, so the consecutive subproduct is a permutation conjugate of

$$P \oplus I_{ab-(a+b-1)} = P \oplus I_{(a-1)(b-1)}.$$

The exact direct-sum law makes its price $p(P)$, even when this padding is empty. Grouping these two calls in the tensor word yields

$$bp(Y_1) + ap(Y_2) = p(Y_1 \otimes Y_2) \leq (b-1)p(Y_1) + (a-1)p(Y_2) + p(P).$$

Thus $p(P) \geq p(Y_1) + p(Y_2)$. Product subadditivity applied to the two identity-padded calls gives the reverse inequality. $\square$

Lemma 7.3 (Scalar pivot invariance). *Under the hypotheses of Theorem 7.1, if $r = 1$, then $p(\widetilde{K}) = p(K)$.*

Proof. Write $D = d \neq 0$. For each positive integer ℓ , use ℓ copies of K , with fresh data groups $x_1, \dots, x_\ell$ of width n and one shared state coordinate. In chronological order the recurrences are

$$y_j = Ax_j + Cw_{j-1}, \quad w_j = Bx_j + dw_{j-1}, \quad 1 \leq j \leq \ell. \quad (7.5)$$

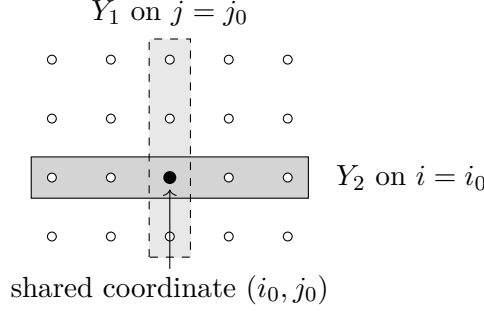


Figure 4: The selected tensor fibers share exactly one coordinate. Their two-call product fixes the $(a-1)(b-1)$ coordinates outside the cross. Other calls of the tensor word occur before or after this consecutive pair.

Let $\mathcal{C}_\ell$ send $(x_1, \dots, x_\ell, w_0)$ to $(y_1, \dots, y_\ell, w_\ell)$. At every extension the previous chain is regarded as a matrix only on its old data groups and its one state coordinate. The next data group is outside this set, so the intersection with the next call is exactly the one state coordinate. Coordinates fixed by the old chain remain members of its old set; no padding from the tensor-grid proof is added to it. Induction using Theorem 7.2 gives

$$p(\mathcal{C}_\ell) = \ell p(K). \quad (7.6)$$

The full chain is invertible as a product of invertible calls. Its state-to-state block is d^ℓ , and its direct block on the data is lower block triangular with diagonal blocks A . In fact its (j, i) block for $i < j$ is $Cd^{j-1-i}B$, and its blocks for $i > j$ vanish. Both blocks required by Theorem 7.1 are therefore invertible for every ℓ .

Pivoting the boundary state means that w_ℓ is given and w_0 is requested. The exact backward recurrence is

$$w_{j-1} = d^{-1}w_j - d^{-1}Bx_j, \quad y_j = (A - Cd^{-1}B)x_j + Cd^{-1}w_j. \quad (7.7)$$

Thus the pivoted chain $\tilde{\mathcal{C}}_\ell$ consists of copies of $\tilde{K}$ in the order $j = \ell, \dots, 1$, again with fresh data groups and one shared state. The same intersection argument gives

$$p(\tilde{\mathcal{C}}_\ell) = \ell p(\tilde{K}). \quad (7.8)$$

Put $S = A - Cd^{-1}B$. Since $\det K = d \det S$, the matrix S is invertible. The reverse chain has state block $d^{-\ell}$ and a direct data block upper triangular in chronological order, with diagonal blocks S ; equivalently it is lower triangular in reverse chronological order. Its full matrix is invertible too. Hence Theorem 7.1 applies both to the forward chain and to the reverse chain. Pivoting the reverse chain returns the forward chain, and therefore

$$\ell p(\tilde{K}) \leq \ell p(K) + 4, \quad \ell p(K) \leq \ell p(\tilde{K}) + 4.$$

Equivalently, $|p(\tilde{K}) - p(K)| \leq 4/\ell$ for every ℓ . Letting ℓ tend to infinity proves equality. The earlier pivot inequality is an exact statement for each finite matrix, so this argument requires no estimate uniform in the increasing data width. $\square$

Corollary 7.4 (Exchange of two basis members). *Let $U = (u_1, \dots, u_n, u_*)$ and $V = (v_1, \dots, v_n, v_*)$ be ordered bases of the same space of linear forms, viewing each list as a column vector of its forms, and let*

$$V = KU, \quad K = \begin{pmatrix} A & C \\ B & d \end{pmatrix}.$$

The exchanged lists

$$U' = (u_1, \dots, u_n, v_*), \quad V' = (v_1, \dots, v_n, u_*)$$

are both bases if and only if $d \neq 0$ and $\det A \neq 0$. Whenever this holds, the transitions $U \mapsto V$ and $U' \mapsto V'$ have equal price. The statement applies to any chosen member of each basis after reordering them to the last positions.

Proof. Relative to U , the coefficient matrices of U' and V' are

$$\begin{pmatrix} I_n & 0 \\ B & d \end{pmatrix}, \quad \begin{pmatrix} A & C \\ 0 & 1 \end{pmatrix},$$

with determinants d and $\det A$, respectively. These give the asserted criterion exactly. When both are nonzero, the transition between them is Equation (7.1) with scalar pivot. Apply Theorem 7.3. Reorderings have zero price. $\square$

7.3 Four forms and the dense-pair bound

Lemma 7.5 (Price of a dense pair). *Every invertible 2×2 matrix all of whose entries are nonzero has price at most $3/2$. In particular,*

$$p\left(\begin{pmatrix} 1 & 1 \\ 1 & u \end{pmatrix}\right) \leq \frac{3}{2} \quad (u \in \mathbb{C} \setminus \{0, 1\}). \quad (7.9)$$

Proof. For a pair matrix H , the tensor law reads $p(H \otimes H) = 4p(H)$. We therefore seek a four-coordinate map of price at most six that becomes a tensor square after monomial changes. Two shears and the basis exchanges below put this map into the form of a rank-one perturbation of the identity, making the required monomial changes explicit.

Fix $\lambda \in \mathbb{C} \setminus \{0, 1\}$ and four independent linear forms a, b, c, d . We first bound the transition from

$$Y^{(0)} = (a, b, c, d) \quad \text{to} \quad X = (a + b, b + c, c + d, d + \lambda a).$$

Its matrix is $I_4 + J$, where

$$J = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ \lambda & 0 & 0 & 0 \end{pmatrix}, \quad J^4 = \lambda I_4.$$

Here J is monomial because $\lambda \neq 0$, and

$$(I_4 + J)(I_4 - J + J^2 - J^3) = (1 - \lambda)I_4$$

shows that $I_4 + J$ is invertible. Apply Theorem 6.2 at parameter one to

$$\mathcal{L} = \begin{pmatrix} I_4 & J \\ I_4 & 0 \end{pmatrix}.$$

This full cell is invertible, its direct block and feedback denominator are identities, and its output is $I_4 + J$. To implement $\mathcal{L}$, send (h, l) to (h, Jl) by a monomial, add h_i into the second coordinate of each of the four pairs, and swap the two blocks. Consequently

$$p(I_4 + J) \leq p(\mathcal{L}) \leq 4. \quad (7.10)$$

We change the first and third members on the Y side using two shears and legal exchanges. For clarity, all intermediate lists are displayed below. Put

$$X_b = (a + b, b, c + d, d + \lambda a), \quad X_d = (a + b, b + c, c + d, d).$$

Every determinant in the last two columns is computed from the coefficient rows in the original ordered basis (a, b, c, d) .

Stage	Y list	X list	$\det Y$	$\det X$
0	(a, b, c, d)	X	1	$1 - \lambda$
1	$(a, b + c, c, d)$	X_b	1	1
2	$(a + b + c, b + c, c, d)$	X_b	1	1
3	$(a + b + c, b, c, d)$	X	1	$1 - \lambda$
4	$(a + b + c, b, c, d + \lambda a)$	X_d	1	1
5	$(a + b + c, b, c + d + \lambda a, d + \lambda a)$	X_d	1	1
6	$(a + b + c, b, c + d + \lambda a, d)$	X	$1 - \lambda$	$1 - \lambda$

The steps $0 \rightarrow 1$ and $2 \rightarrow 3$ exchange the second members; the steps $3 \rightarrow 4$ and $5 \rightarrow 6$ exchange the fourth members. The step $1 \rightarrow 2$ adds the second member of Y into its first, and $4 \rightarrow 5$ adds its fourth member into its third. These are the only two shears. The determinant table proves that every list is a basis, including on both sides of every exchange; it follows also by direct elementary row operations. Thus Theorem 7.4 applies at all four exchanges. Changing the source basis from Y to EY by a shear replaces its transition matrix T by TE^{-1} , increasing price by at most one. Together with Equation (7.10), this proves that the transition from the last Y list to X has price at most six.

Reorder these final lists as

$$\bar{Y} = (a + b + c, b, d, c + d + \lambda a), \quad \bar{X} = (b + c, b + a, d + \lambda a, c + d).$$

Let

$$k = (1, -1, 1, -1)^\top, \quad \Delta = (-1, 1, \lambda, -\lambda)^\top.$$

Then $k^\top \bar{Y} = (1 - \lambda)a$ and $\bar{X} - \bar{Y} = \Delta a$, so the transition matrix is the explicit rank-one perturbation

$$T = I_4 + \frac{\Delta k^\top}{1 - \lambda}, \quad p(T) \leq 6. \quad (7.11)$$

We now give the diagonal scalings. All their entries are nonzero under the stated hypotheses on λ . Set $u = \lambda^{-1}$ and

$$L_1 = \text{diag}\left(\frac{1 - \lambda}{\Delta_1}, \frac{1 - \lambda}{\Delta_2}, \frac{1 - \lambda}{\Delta_3}, \frac{1 - \lambda}{\Delta_4}\right), \quad R_1 = \text{diag}(1, -1, 1, -1).$$

The rank-one summand of $L_1 T R_1$ has every entry equal to one, and a direct calculation gives

$$L_1 T R_1 = \begin{pmatrix} \lambda & 1 & 1 & 1 \\ 1 & \lambda & 1 & 1 \\ 1 & 1 & u & 1 \\ 1 & 1 & 1 & u \end{pmatrix}.$$

With the further scalings

$$L_2 = \text{diag}(u, u, u, 1), \quad R_2 = \text{diag}(1, 1, 1, \lambda),$$

we obtain

$$L_2 L_1 T R_1 R_2 = \begin{pmatrix} 1 & u & u & 1 \\ u & 1 & u & 1 \\ u & u & u^2 & 1 \\ 1 & 1 & 1 & 1 \end{pmatrix}. \quad (7.12)$$

Write $B(u) = \begin{pmatrix} 1 & 1 \\ 1 & u \end{pmatrix}$. The (α, γ) entry of $B(u) \otimes B(u)$, for bit pairs $\alpha, \gamma \in \{0, 1\}^2$, is $u^{\alpha_1 \gamma_1 + \alpha_2 \gamma_2}$. Ordering its rows by 10, 01, 11, 00 and its columns by 01, 10, 11, 00 gives exactly Equation (7.12). All the scalings and permutations preserve price. The tensor law therefore yields

$$4p(B(u)) = p(B(u) \otimes B(u)) = p(T) \leq 6.$$

Every $u \neq 0, 1$ arises from the permitted choice $\lambda = 1/u$, proving Equation (7.9).

Finally, if $H = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ is dense and invertible, then

$$\text{diag}(a^{-1}, c^{-1}) H \text{diag}(1, a/b) = B\left(\frac{ad}{bc}\right).$$

Density makes $ad/(bc) \neq 0$, and invertibility makes it different from one. Monomial invariance completes the proof. $\square$

8 Signed matrices and the reflection contradiction

We continue under the assumption that there is no finite win. Thus the price function of Theorem 4.1, the feedback and specialization laws, and the pivot laws all apply. We construct a real involution with a large price and then use the orthogonal reflection in its positive eigenspace to obtain incompatible price bounds. All dimensions in the two specializations below are fixed before either specialization is made.

8.1 Signed bit operators with an exact price

On one bit, set

$$C = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

For $m \geq 1$, define matrices on m bits by

$$D_1 = C, \quad D_{m+1} = C \otimes I_{2^m} + Z \otimes D_m = \begin{pmatrix} D_m & 0 \\ I_{2^m} & -D_m \end{pmatrix}, \quad T_m = I_{2^m} + D_m. \quad (8.1)$$

The matrices D_m are strictly lower triangular in lexicographic bit order. Squaring the displayed block matrix shows inductively that $D_m^2 = 0$. In particular, T_m is invertible and $T_m^{-1} = I_{2^m} - D_m$.

The signed tensor summands in this recursion have the standard Jordan–Wigner creation-operator form, with tensor factors in the reverse order of the convention in Seeley et al. [18, Section II.B, Equations (10)–(11)]. Here the bits label ordinary matrix coordinates. The recursion above and the price arguments below provide all identities needed for the proof.

Lemma 8.1. *For every integer $m \geq 1$,*

$$p(T_m) = m2^{m-1}.$$

Proof. We have $T_1 = U$, so $p(T_1) = 1$. Put $n = 2^m$ and consider the transition from input forms $x = (x_1, \dots, x_{2n})$ to output forms $y = K_0 x$, where

$$K_0 = U \otimes T_m = \begin{pmatrix} T_m & 0 \\ T_m & T_m \end{pmatrix}.$$

We shall exchange each bottom input form with the corresponding bottom output form, using only the exact *scalar* pivot law of Theorem 7.3. Here is a determinant check for every intermediate exchange.

For any subset $H \subseteq \{n+1, \dots, 2n\}$ of already exchanged positions, keep the forms in their original index positions and set

$$u_i = \begin{cases} y_i & i \in H, \\ x_i & i \notin H, \end{cases} \quad v_i = \begin{cases} x_i & i \in H, \\ y_i & i \notin H. \end{cases}$$

Let U_H, V_H be their coefficient matrices relative to x . Simultaneously permuting rows and columns to put H first gives

$$U_H \sim \begin{pmatrix} K_0[H, H] & K_0[H, H^c] \\ 0 & I \end{pmatrix}, \quad V_H \sim \begin{pmatrix} I & 0 \\ K_0[H^c, H] & K_0[H^c, H^c] \end{pmatrix}.$$

The row and column permutation signs cancel. Every principal submatrix of the unit lower triangular matrix K_0 is unit lower triangular, so

$$\det U_H = \det K_0[H, H] = 1, \quad \det V_H = \det K_0[H^c, H^c] = 1. \quad (8.2)$$

The empty principal minor, when it occurs, has determinant one. Thus both lists are bases and their current transition $Q_H = V_H U_H^{-1}$ has determinant one.

For the next index $j \notin H$ in the bottom half, replacing u_j by v_j changes the input determinant by the factor $(Q_H)_{jj}$. Replacing v_j by u_j changes the output determinant by $(Q_H^{-1})_{jj}$. Applying (8.2) to H and $H \cup \{j\}$ therefore gives

$$(Q_H)_{jj} = \frac{\det U_{H \cup \{j\}}}{\det U_H} = 1, \quad (Q_H^{-1})_{jj} = \frac{\det V_{H \cup \{j\}}}{\det V_H} = 1.$$

Move position j to the last port in both bases. The scalar diagonal block required for the pivot is consequently one; the determinant of its complementary principal block is, by the cofactor identity,

$$\det Q_H (Q_H^{-1})_{jj} = 1.$$

Both hypotheses of Theorem 7.3 hold at every step, and each exchange preserves the price exactly.

After all bottom positions have been exchanged, write the original inputs as (a, b) and the bottom output as $z = T_m a + T_m b$. Solving gives $b = T_m^{-1} z - a$. Thus the final transition, from (a, z) to $(T_m a, b)$, is

$$\begin{pmatrix} T_m & 0 \\ -I & T_m^{-1} \end{pmatrix}.$$

Conjugating by $\text{diag}(I, -I)$ changes the lower left block to I ; the result is T_{m+1} by (8.1). Monomial invariance and the tensor law now give

$$p(T_{m+1}) = p(U \otimes T_m) = 2^m + 2p(T_m),$$

which proves the formula by induction. □

Fix an integer $d \geq 1$. On $d + 1$ bits, of width $N = 2^{d+1}$, put

$$\Gamma = X \otimes I_{2^d} + Z \otimes D_d = \begin{pmatrix} D_d & I \\ I & -D_d \end{pmatrix}. \quad (8.3)$$

Since $D_d^2 = 0$, this is a real involution. Its price is also exact:

$$p(\Gamma) = \frac{dN}{2}. \quad (8.4)$$

To verify this, multiplication on the left by the monomial $X \otimes I$ gives $I + (XZ) \otimes D_d$. The recursion expands as

$$D_d = \sum_{i=1}^d Z^{\otimes(i-1)} \otimes C \otimes I_{2^{d-i}}.$$

Each nonzero entry therefore changes exactly one bit from zero to one, increasing Hamming weight by one. Define the signed monomial P to act on the first bit by $(XZ)^{-|b|}$ when the last d bits have value b and Hamming weight $|b|$. On a transition of these last bits from weight j to weight $j + 1$, conjugation by P changes the first-bit factor to

$$(XZ)^{-(j+1)}(XZ)(XZ)^j = I_2.$$

It leaves the identity term unchanged. Hence

$$P(I + (XZ) \otimes D_d)P^{-1} = I_2 \otimes T_d,$$

and (8.4) follows from Theorem 8.1 and the tensor law.

We next record a transpose symmetry. Define the signed orthogonal monomial

$$J = \bigotimes_{i=1}^{d+1} (Z^{i-1} X). \quad (8.5)$$

For its i th factor $H_i = Z^{i-1} X$, direct multiplication gives

$$H_i^\top = (-1)^{i-1} H_i, \quad H_i Z H_i^{-1} = -Z, \quad H_i C H_i^{-1} = (-1)^{i-1} C^\top.$$

The first-bit term of Γ is $X \otimes I$ and is fixed by this conjugation. Every other term has the form

$$Z^{\otimes(i-1)} \otimes C \otimes I_{2^{d+1-i}}, \quad 2 \leq i \leq d+1.$$

The $i - 1$ prefix signs and the sign from the C factor cancel. It follows that

$$J\Gamma J^{-1} = \Gamma^\top, \quad J^\top = (-1)^{d(d+1)/2} J. \quad (8.6)$$

On width $w = N^2$, set

$$M = \Gamma \otimes \Gamma, \quad L = \Gamma \otimes I_N - I_N \otimes \Gamma, \quad S = J \otimes J, \quad A = SL. \quad (8.7)$$

The matrix S is a symmetric orthogonal monomial, so $S^{-1} = S$. Equation (8.6) gives $SL S^{-1} = L^\top$, and consequently

$$A^\top = L^\top S^\top = SL = A.$$

In particular, A is real symmetric. Moreover,

$$(\Gamma \otimes I_N)L = I_w - M, \quad E := \ker A = \ker L = \ker(I_w - M). \quad (8.8)$$

Thus E is the positive eigenspace of the involution M .

We shall need a count of the supported unordered off-diagonal pairs of A . The first-bit flip in Γ has N nonzero entries, and each of its other d bit terms has $N/2$ nonzero entries. Their supports are disjoint, since they change different bits. Thus $\text{nnz}(\Gamma) = (1 + d/2)N$. The supports of $\Gamma \otimes I_N$ and $I_N \otimes \Gamma$ are also disjoint: they change a bit in different tensor factors. Therefore

$$\text{nnz}(L) = (d + 2)w, \quad \text{nnz}(A) = (d + 2)w.$$

Symmetry makes every supported off-diagonal pair account for two entries. If h is the number of such pairs, we have

$$h \leq (d/2 + 1)w. \quad (8.9)$$

Let R be the orthogonal reflection in E . The rest of the argument will establish the bounds

$$(d - 4)w \leq p(R) \leq \left(\frac{3d}{4} + \frac{7}{2}\right)w.$$

They are incompatible when $d > 30$. The upper bound uses the description $E = \ker A$: a polynomial perturbation has one dense pair factor of price at most $3/2$ for each supported pair of A , at generic parameter values. Two algebraic specializations turn this bound into a price for the reflection. The lower bound uses E as the positive eigenspace of M , whose tensor price is dw . Its graph description will recover the full price of M from R , at a cost of at most $4w$. Thus the contradiction comes from the different coefficients of d , while the changes of boundary coordinates cost only a fixed multiple of the width.

8.2 Pricing a reflection by two algebraic specializations

The following statement isolates the use of sparsity. Orthogonality refers to the usual real inner product; all real maps and subspaces are then complexified when their prices are taken.

Lemma 8.2. *Let A be a real symmetric $w \times w$ matrix with h supported unordered off-diagonal pairs. Let R act as $+I$ on $\ker A$ and as $-I$ on $(\ker A)^\perp$. Then*

$$p(R) \leq \frac{3}{2}h + 2w.$$

Proof. Write e_{ij} for matrix units, and fix any order of the supported pairs. Form the polynomial matrix

$$K(\varepsilon) = (I + \varepsilon \text{diag}(A_{11}, \dots, A_{ww})) \prod_{\substack{i < j \\ A_{ij} \neq 0}} (I + \varepsilon A_{ij}(e_{ij} + e_{ji})). \quad (8.10)$$

Then $K(0) = I$ and $K'(0) = A$. Outside a finite set E_0 , the diagonal factor is an invertible monomial and every other factor acts on its two coordinates as the dense invertible matrix

$$\begin{pmatrix} 1 & \varepsilon A_{ij} \\ \varepsilon A_{ij} & 1 \end{pmatrix}.$$

For example, E_0 may contain zero, the roots of the diagonal-factor determinant, and all roots of $1 - \varepsilon^2 A_{ij}^2$ for the supported pairs. These are finitely many roots of nonzero polynomials. By Theorem 7.5, direct sums, and product subadditivity,

$$p(K(\varepsilon)) \leq \frac{3}{2}h \quad (\varepsilon \notin E_0). \quad (8.11)$$

Put $Q(\varepsilon) = (K(\varepsilon) - I)/\varepsilon$, with its polynomial extension $Q(0) = A$. First fix a parameter t outside the single finite set

$$\mathcal{E} = \{0\} \cup \{\lambda^{-1}, -\lambda^{-1} : \lambda \in \text{spec}(A), \lambda \neq 0\}. \quad (8.12)$$

The polynomial matrices

$$C_+(\varepsilon) = I + tQ(\varepsilon), \quad C_-(\varepsilon) = I - tQ(\varepsilon)$$

have nonzero determinants at zero, since $I \pm tA$ are invertible. Consequently the rational output

$$H_t(\varepsilon) = C_+(\varepsilon)C_-(\varepsilon)^{-1}$$

is regular and invertible at zero.

For this fixed t , exclude also $\varepsilon = 0, t, -t$ and the roots of $\det C_+(\varepsilon) \det C_-(\varepsilon)$, in addition to E_0 . This defines a finite set E_t , which may depend on t . For $\varepsilon \notin E_t$ define

$$m = \frac{\varepsilon - t}{\varepsilon + t}, \quad s = \frac{t}{\varepsilon + t}, \quad v = \frac{2\varepsilon t}{(\varepsilon + t)^2},$$

and consider the data-state cell

$$F_{t,\varepsilon} = \begin{pmatrix} mI_w & K(\varepsilon) \\ vI_w & sK(\varepsilon) \end{pmatrix} = \begin{pmatrix} mI_w & I_w \\ vI_w & sI_w \end{pmatrix} \text{diag}(I_w, K(\varepsilon)). \quad (8.13)$$

The first factor is a direct sum, after reordering, of w scalar pair matrices. Their determinants are

$$ms - v = -\frac{t}{\varepsilon + t} \neq 0.$$

Hence the cell is invertible. Its direct block mI_w is invertible, and its feedback denominator at feedback parameter one satisfies

$$I_w - sK(\varepsilon) = \frac{\varepsilon}{\varepsilon + t}C_-(\varepsilon), \quad (8.14)$$

so that denominator is invertible too. Its feedback output is exactly

$$\begin{aligned} mI_w + K(\varepsilon)(I_w - sK(\varepsilon))^{-1}v &= (mI_w + (v - ms)K(\varepsilon))(I_w - sK(\varepsilon))^{-1} \\ &= C_+(\varepsilon)C_-(\varepsilon)^{-1} = H_t(\varepsilon). \end{aligned}$$

The last equality follows from $mI_w + (v - ms)K(\varepsilon) = \varepsilon C_+(\varepsilon)/(\varepsilon + t)$. The output is invertible by the choice of E_t , so every hypothesis of Theorem 6.2 is satisfied.

An invertible scalar pair has price at most two. The factorization (8.13) and (8.11) therefore imply

$$p(H_t(\varepsilon)) \leq p(F_{t,\varepsilon}) \leq p(K(\varepsilon)) + 2w \leq \frac{3}{2}h + 2w \quad (\varepsilon \notin E_t).$$

Apply Theorem 6.3 to the rational matrix H_t at $\varepsilon = 0$. This proves

$$p((I + tA)(I - tA)^{-1}) \leq C_A, \quad C_A = \frac{3}{2}h + 2w, \quad t \notin \mathcal{E}. \quad (8.15)$$

The quantifiers here are useful: A, w, h and C_A were fixed first; for *each* $t \notin \mathcal{E}$ there is a finite E_t giving the same bound, and specialization is applied separately for that t . Thus (8.15) holds for every t outside one finite set, with one price bound. Only the rational output is specialized. At $\varepsilon = 0$ the cell itself would have $m = -1$, $s = 1$, $v = 0$, and $K = I$, making its feedback denominator zero; no feedback law is applied to that cell.

For the second specialization use $z = 1/t$ and the rational matrix

$$Y(z) = (zI_w + A)(zI_w - A)^{-1}.$$

By the Real Spectral Theorem, $A = O \operatorname{diag}(\lambda_1, \dots, \lambda_w) O^T$ for a real orthogonal O and real eigenvalues λ_i . In this basis, the corresponding rational functions are $(z + \lambda_i)/(z - \lambda_i)$. If $\lambda_i = 0$, this function is identically one after cancellation; if $\lambda_i \neq 0$, it is regular at zero with value minus one. It follows that Y is regular at zero and $Y(0) = R$, an invertible matrix. In particular, the zero eigenspace is semisimple; real symmetry supplies exactly the diagonalizability needed for this removable-singularity assertion.

For all z outside $\{0\} \cup \{\lambda, -\lambda : \lambda \in \operatorname{spec}(A), \lambda \neq 0\}$, we may take $t = 1/z$ in (8.15). Hence $p(Y(z)) \leq C_A$ outside a finite set. A second application of Theorem 6.3 yields $p(R) \leq C_A$. The orthogonal diagonalization was used to identify a rational matrix and its value; it was not used as a priced implementation. Neither specialization assumes continuity of p . $\square$

Apply Theorem 8.2 to $A = SL$ from (8.7). With R the orthogonal reflection in E from (8.8), (8.9) gives

$$p(R) \leq \frac{3}{2}(d/2 + 1)w + 2w = \left(\frac{3d}{4} + \frac{7}{2}\right)w. \quad (8.16)$$

8.3 The graph of the positive eigenspace

Every nonzero term of Γ changes exactly one bit, so Γ anticommutes with the parity sign $\Pi = Z^{\otimes(d+1)}$. Consequently M anticommutes with $\Pi \otimes I_N$. Order the coordinates according to this first-factor parity. The two parts have equal width $r = w/2$, and in this order

$$M = \begin{pmatrix} 0 & B \\ B^{-1} & 0 \end{pmatrix} \quad \text{for some } B \in \operatorname{GL}_r(\mathbb{R}). \quad (8.17)$$

Indeed anticommutation forces the diagonal blocks to vanish, and $M^2 = I$ forces the two off-diagonal blocks to be mutual inverses. The tensor law and (8.4) give

$$p(M) = 2Np(\Gamma) = dw = 2p(B), \quad (8.18)$$

where the last equality follows by multiplying (8.17) by a block-swap permutation and then using direct sums and inverse symmetry.

Equation (8.17) identifies

$$E = \{(Bv, v) : v \in \mathbb{R}^r\}, \quad E^\perp = \{(u, v) : B^T u + v = 0\}.$$

Let $Q = I_r + B^\top B$. For every nonzero real vector v , $v^\top Q v = \|v\|^2 + \|Bv\|^2 > 0$; hence Q is invertible. The columns of $\begin{pmatrix} B \\ I_r \end{pmatrix}$ form a basis of E , so its orthogonal projection is

$$P_E = \begin{pmatrix} B \\ I_r \end{pmatrix} Q^{-1} \begin{pmatrix} B^\top & I_r \end{pmatrix}.$$

As $R = 2P_E - I_w$, its off-diagonal blocks in the parity order are

$$R_{+,-} = 2BQ^{-1}, \quad R_{-,+} = 2Q^{-1}B^\top = R_{+,-}^\top. \quad (8.19)$$

Both are invertible. This use of positive definiteness is over $\mathbb{R}$; all these invertible real matrices also define invertible complex linear maps.

For $y = Rx$, order the input as (x_-, x_+) and the output as (y_+, y_-) . The resulting transition is

$$K_R = \begin{pmatrix} R_{+,-} & R_{+,+} \\ R_{-,-} & R_{-,+} \end{pmatrix}.$$

The two diagonal blocks required by Theorem 7.1 are now precisely the invertible blocks in (8.19). Pivot the second group of r ports and then swap the two output groups. The resulting map

$$V : (x_-, y_-) \mapsto (x_+, y_+)$$

therefore satisfies

$$p(V) \leq p(R) + 4r. \quad (8.20)$$

Since R is the reflection in E , the vectors $x + y$ and $x - y$ belong to E and $E^\perp$, respectively. Their graph equations are

$$x_+ + y_+ = B(x_- + y_-), \quad x_+ - y_+ = -B^{-\top}(x_- - y_-).$$

For the invertible sum-and-difference map

$$F = \begin{pmatrix} I_r & I_r \\ I_r & -I_r \end{pmatrix}, \quad F^{-1} = \frac{1}{2}F,$$

these equations state the exact identity

$$FVF^{-1} = \text{diag}(B, -B^{-\top}). \quad (8.21)$$

The relation between the coordinates is displayed in Figure 5.

After interleaving coordinates, F is a direct sum of r invertible scalar pair maps. Thus $p(F) = p(F^{-1}) \leq 2r$. Combining (8.18), (8.20), and (8.21), and using transpose and inverse symmetry, yields

$$dw = 2p(B) = p(\text{diag}(B, -B^{-\top})) \leq p(V) + 4r \leq p(R) + 8r = p(R) + 4w. \quad (8.22)$$

Theorem 8.3 (Existence of a finite win). *There exist an invertible nonmonomial matrix A of a finite width q , an integer $b \geq 2$, and a word on exactly q^b coordinates implementing $A^{\otimes b}$ with monomials and fewer than bq^{b-1} calls to A .*

$$\begin{array}{ccc}
(x_-, y_-) & \xrightarrow{V} & (x_+, y_+) \\
F \downarrow & & \downarrow F \\
(x_- + y_-, x_- - y_-) & \xrightarrow{\text{diag}(B, -B^{-\top})} & (x_+ + y_+, x_+ - y_+)
\end{array}$$

Figure 5: Pivoting the reflection exposes its two graph equations. The sums are the parity components of $x + y \in E$; the differences are the parity components of $x - y \in E^\perp$. All four maps are exact invertible linear maps.

Proof. If there were no finite win, the preceding construction and all its price bounds would apply for every fixed integer $d \geq 1$. Equations (8.16) and (8.22) would imply

$$dw \leq \left(\frac{3d}{4} + \frac{15}{2} \right) w.$$

Since $w > 0$, this says $d \leq 30$, contrary to choosing any fixed integer $d > 30$. This contradiction proves the theorem. $\square$

By Theorem 3.4, the finite win of Theorem 8.3 supplies lengths $n_j \rightarrow \infty$ and exact circuits in the stated scalar-gate model whose sizes s_j satisfy

$$\frac{s_j}{n_j \log_2 n_j} \rightarrow 0.$$

For every $c > 0$ and every integer cutoff $n_0 \geq 2$, some j therefore has $n_j \geq n_0$ and $s_j < cn_j \log_2 n_j$. This proves the precise negative alternative in Theorem 1.1. The conclusion is nonuniform existence along an unbounded sequence; the auxiliary price zero of monomials and the two specializations do not alter the scalar gate costs or the exactness of those circuits.

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